Mini project report on

DISASTER MANAGEMENT SYSTEM

Submitted in partial fulfilment of the requirements for the award of degree of

Bachelor of Technology

in

Computer Science & Engineering

UE24CS351A – DBMS Project

Submitted by:

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AUG - DEC 2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FACULTY OF ENGINEERING

PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)

Electronic City, Hosur Road, Bengaluru – 560 100, Karnataka, India

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CERTIFICATE

This is to certify that the mini project entitled

Disaster Management System

is a bonafide work carried out by

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In partial fulfilment for the completion of fifth semester DBMS Project (UE24CS351A) in the Program of Study -Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period AUG. 2026 – DEC. 2026. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report. The project has been approved as it satisfies the 5th semester academic requirements in respect of project work.

Signature
Dr. Shilpa S
Assistant Professor

DECLARATION

We hereby declare that the DBMS Project entitled **Disaster Management System** has been carried out by us under the guidance of **Dr. Shilpa S, Assistant Professor** and submitted in partial fulfilment of the course requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester AUG – DEC 2026.

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ABSTRACT

Disaster relief teams have to track affected people, shelters, supply depots, stock levels and delivery vehicles at the same time, and usually do it with scattered spreadsheets. The Disaster Management System is a database-driven application that keeps all of this in one place. Structured data (victims, depots, shelters, routes, supplies, vehicles, requests, assignments and audit history) is stored in MySQL. Semi-structured data (incident reports and field updates with varying fields and GeoJSON locations) is stored in MongoDB. Business rules are enforced inside the database through CHECK and foreign-key constraints, triggers, functions and stored procedures. These rules cover stock deduction, nearest-supply and nearest-vehicle matching, priority-based allocation and automatic status logging. A Node.js/Express REST API with JWT authentication and three roles (admin, logistics manager, field worker) sits on top. It is covered by 306 automated checks and a 45-check live smoke test.

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1. INTRODUCTION

1.1 Overview.
During floods, landslides and similar emergencies, relief organisations must decide quickly who needs what, where the stock is, and which vehicle should carry it. Delays and double allocation of scarce supplies cost lives. This project builds a database-centred system that records relief resources and automates their allocation.

1.2 Why two databases.

- MySQL (relational) holds data with fixed structure and strong integrity needs. Examples are stock quantities, which must never go negative, and request → victim relationships.
- MongoDB (document) holds incident reports and field updates. Their fields vary (a landslide report differs from a drought report), they contain arrays such as additional_requirements, and they need geospatial queries.

1.3 Technologies used.

Layer	Technology
Relational database	MySQL 8.0.16+ (needed for enforced CHECK constraints)
Document database	MongoDB 7 (run in Docker)
Backend	Node.js 20+, Express, JWT authentication
Testing	Custom test harness (npm test) and live smoke test (npm run smoke)
Version control	Git and GitHub

1.4 Scope.
The system manages victims, depots, shelters, routes, supplies, vehicles and resource requests, and automatically matches requests to supplies. It also records every request status change, stores incident reports and field updates, and provides dashboard aggregates.

2. PROBLEM DEFINITION

2.1 Problem statement.
Relief operations lack a single, consistent data store. Stock is tracked on paper or in spreadsheets, so supplies get double-booked, critical victims wait behind low-priority ones, vehicles are assigned without regard to distance, and field observations are not linked to the requests they relate to.

2.2 Objectives.

1. Store victims with location, group size and priority (critical/high/medium/low).
2. Maintain depots and shelters (with capacity and occupancy), road routes with open/blocked status, supplies and vehicles.
3. Let field workers raise resource requests and track them through pending → matched → delivered or cancelled.
4. Automatically match requests to the nearest sufficient supply and nearest suitable vehicle, serving critical and larger groups first.
5. Guarantee integrity at database level: no negative stock, no shelter over capacity, no wrong-resource assignment.
6. Keep an automatic audit trail of every request status change.
7. Store flexible incident reports and field updates in MongoDB, with geospatial search, and link them to MySQL requests.
8. Restrict actions by role.

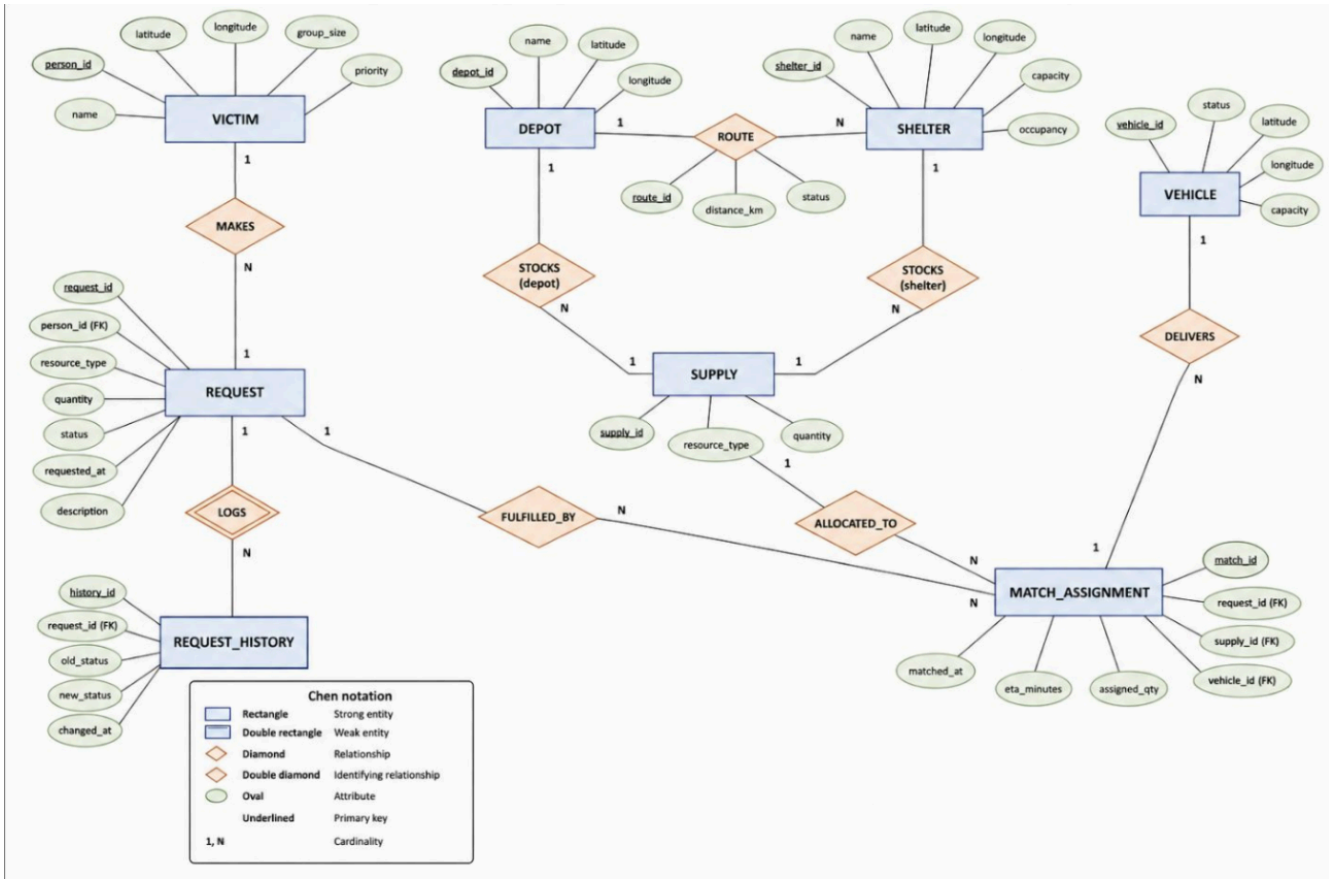
2.3 Functional requirements.

- Authenticated login with three roles (admin, logistics_manager, field_worker); passwords stored only as bcrypt hashes.
- CRUD for victims, depots, shelters, supplies, vehicles and routes.
- Request creation (with a warning if no single supply can currently cover it) and cancellation of pending requests.
- Single-request matching, bulk matching of all pending requests, and delivery completion.
- Dashboard: counts by status/priority, shelter occupancy, supply-vs-demand, priority queue.
- Incident and field-update management, nearest-incident and nearest-depot lookups.

2.4 Non-functional requirements.

- Integrity
- Concurrency safety (matching runs under a MySQL named lock so two operations cannot assign the same stock)
- Security (JWT, role checks, hashed passwords)
- Testability

3. ER MODEL


VICTIM
person_id
name
latitude
longitude
group_size
priority
MAKES
REQUEST
request_id
person_id (FK)
resource_type
quantity
status
requested_at
description
LOGS
REQUEST_HISTORY
history_id
request_id (FK)
old_status
new_status
changed_at
DEPOT
depot_id
name
latitude
longitude
ROUTE
route_id
distance_km
status
SHELTER
shelter_id
name
latitude
longitude
capacity
occupancy
STOCKS (depot)
STOCKS (shelter)
SUPPLY
supply_id
resource_type
quantity
FULFILLED_BY
ALLOCATED_TO
VEHICLE
vehicle_id
status
latitude
longitude
capacity
DELIVERS
MATCH_ASSIGNMENT
match_id
request_id (FK)
supply_id (FK)
vehicle_id (FK)
matched_at
eta_minutes
assigned_qty
Chen notation
Rectangle Strong entity
Double rectangle Weak entity
Diamond Relationship
Double diamond Identifying relationship
Oval Attribute
Underlined Primary key
1, N Cardinality

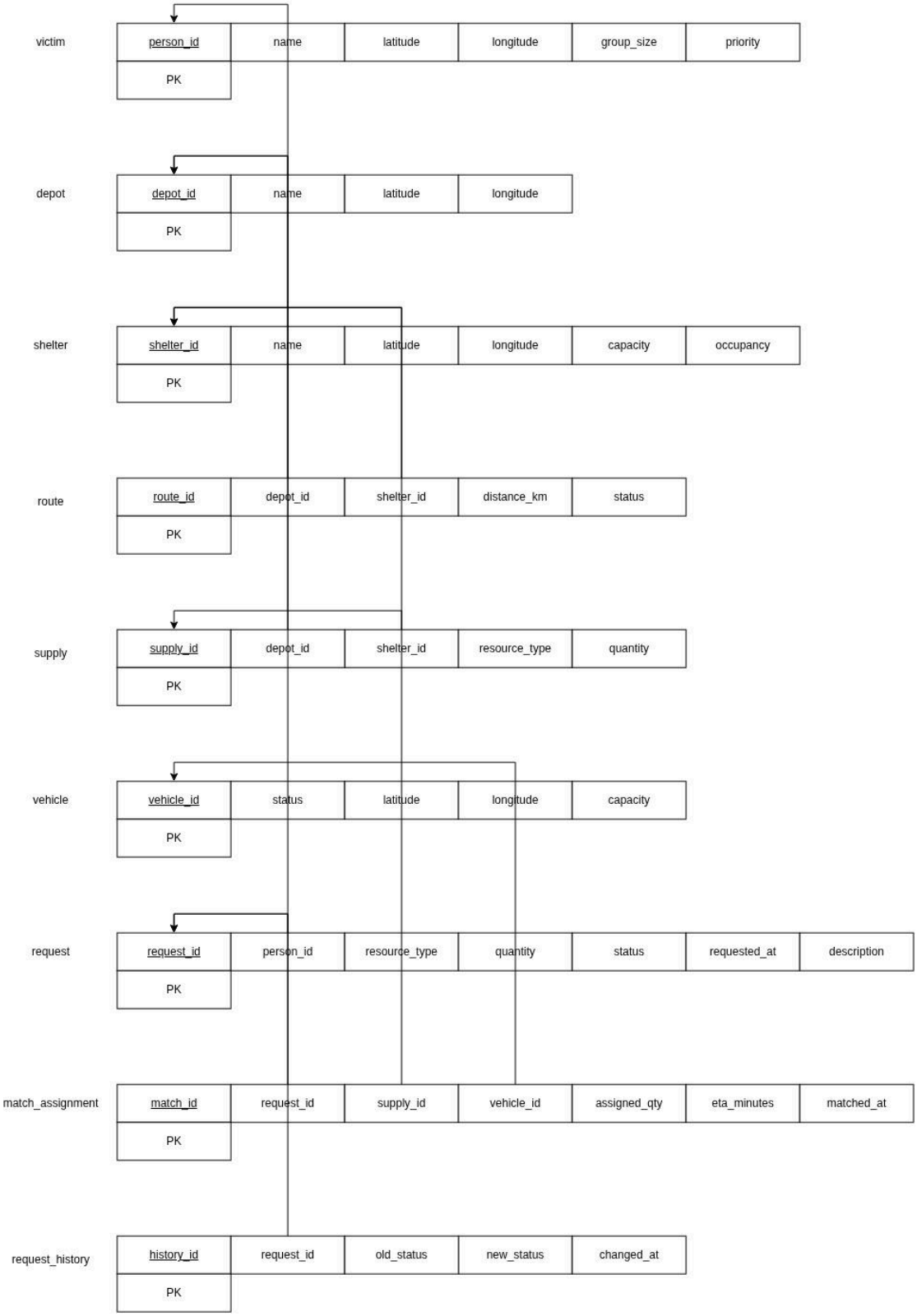
4. ER TO RELATIONAL MAPPING

4.1 STEPS OF ALGORITHM FOR CHOSEN PROBLEM

1. Strong entities. VICTIM, DEPOT, SHELTER, SUPPLY, VEHICLE, REQUEST, USERS each become a table with their simple attributes and primary key.
2. Weak entities. None. Every entity has its own key.
3. 1:1 relationships. REQUEST–MATCH_ASSIGNMENT: match_assignment takes request_id as a foreign key with a UNIQUE constraint, which enforces one match per request.
4. 1:N relationships.
    - request.person_id → victim
    - match_assignment.supply_id → supply
    - match_assignment.vehicle_id → vehicle (nullable)
    - request_history.request_id → request
    - supply.depot_id → depot and supply.shelter_id → shelter, with a CHECK ensuring exactly one is non-null (exclusive arc)
5. M:N relationships. DEPOT–SHELTER becomes the table route(depot_id, shelter_id, distance_km, status). The INCIDENT–REQUEST link lives in MongoDB as the array related_request_ids.
6. Multivalued attributes. None in MySQL. In MongoDB, additional_requirements is stored as an array, which avoids a separate table.
7. N-ary relationships. None.
8. Integrity rules added at mapping time: ENUM columns for fixed vocabularies, CHECKs for non-negative stock and valid occupancy, and UNIQUE (location, resource_type) so each location has one stock row per resource.

Normalisation: every table is in 3NF. Non-key attributes depend only on the key, and the audit data is split into request_history.

4.2 COMPLETE DIAGRAM OF RELATIONAL MAPPING



5. DDL STATEMENTS

STATEMENTS WITH SCREEN SHOTS OF THE TABLE CREATION

```
(PES2UG24CS655@root@localhost) [(none)] >  CREATE DATABASE disaster;
Query OK, 1 row affected (0.02 sec)

(PES2UG24CS655@root@localhost) [(none)] >  USE disaster;
Database changed
(PES2UG24CS655@root@localhost) [disaster] >  CREATE TABLE victim (
    ->      person_id    INT PRIMARY KEY AUTO_INCREMENT,
    ->      name         VARCHAR(150),
    ->      latitude     DECIMAL(9,6) NOT NULL,
    ->      longitude    DECIMAL(9,6) NOT NULL,
    ->      group_size   INT DEFAULT 1,
    ->      priority     ENUM('critical','high','medium','low') DEFAULT 'medium'
    -> );
Query OK, 0 rows affected (0.03 sec)

(PES2UG24CS655@root@localhost) [disaster] >
(PES2UG24CS655@root@localhost) [disaster] >  CREATE TABLE depot (
    ->      depot_id     INT PRIMARY KEY AUTO_INCREMENT,
    ->      name         VARCHAR(150) NOT NULL,
    ->      latitude     DECIMAL(9,6) NOT NULL,
    ->      longitude    DECIMAL(9,6) NOT NULL
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
(PES2UG24CS655@root@localhost) [disaster] >  CREATE TABLE shelter (
    ->      shelter_id   INT PRIMARY KEY AUTO_INCREMENT,
    ->      name         VARCHAR(150) NOT NULL,
    ->      latitude     DECIMAL(9,6) NOT NULL,
    ->      longitude    DECIMAL(9,6) NOT NULL,
    ->      capacity     INT NOT NULL,
    ->      occupancy    INT DEFAULT 0,
    ->      CHECK (occupancy >= 0 AND occupancy <= capacity)
    -> );
Query OK, 0 rows affected (0.03 sec)

(PES2UG24CS655@root@localhost) [disaster] >  CREATE TABLE route (
    ->      route_id     INT PRIMARY KEY AUTO_INCREMENT,
    ->      depot_id     INT NOT NULL,
    ->      shelter_id   INT NOT NULL,
    ->      distance_km  DECIMAL(6,2),
    ->      status       ENUM('open','blocked') DEFAULT 'open',
    ->      FOREIGN KEY (depot_id) REFERENCES depot(depot_id),
    ->      FOREIGN KEY (shelter_id) REFERENCES shelter(shelter_id)
    -> );
Query OK, 0 rows affected (0.03 sec)

(PES2UG24CS655@root@localhost) [disaster] >  CREATE TABLE supply (
    ->      supply_id     INT PRIMARY KEY AUTO_INCREMENT,
    ->      depot_id      INT,
    ->      shelter_id    INT,
    ->      resource_type VARCHAR(50) NOT NULL,
    ->      quantity      INT NOT NULL,
    ->      CHECK (quantity >= 0),
    ->      CHECK ((shelter_id IS NOT NULL AND depot_id IS NULL)
    ->          OR (shelter_id IS NULL AND depot_id IS NOT NULL)),
    ->      FOREIGN KEY (depot_id) REFERENCES depot(depot_id),
    ->      FOREIGN KEY (shelter_id) REFERENCES shelter(shelter_id)
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
(PES2UG24CS655@root@localhost) [disaster] >  CREATE TABLE vehicle (
    ->     vehicle_id   INT PRIMARY KEY AUTO_INCREMENT,
    ->     status       ENUM('available','en_route','maintenance') DEFAULT 'available',
    ->     latitude     DECIMAL(9,6),
    ->     longitude    DECIMAL(9,6),
    ->     capacity     INT NOT NULL
    -> );
Query OK, 0 rows affected (0.02 sec)

(PES2UG24CS655@root@localhost) [disaster] >  CREATE TABLE request (
    ->     request_id    INT PRIMARY KEY AUTO_INCREMENT,
    ->     person_id     INT NOT NULL,
    ->     resource_type VARCHAR(50) NOT NULL,
    ->     quantity      INT NOT NULL CHECK (quantity > 0),
    ->     status        ENUM('pending','matched','delivered','cancelled') DEFAULT 'pending',
    ->     requested_at  TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     description TEXT,
    ->     FOREIGN KEY (person_id) REFERENCES victim(person_id)
    -> );
Query OK, 0 rows affected (0.03 sec)

(PES2UG24CS655@root@localhost) [disaster] >  CREATE TABLE match_assignment (
    ->     match_id     INT PRIMARY KEY AUTO_INCREMENT,
    ->     request_id   INT NOT NULL UNIQUE,
    ->     supply_id    INT NOT NULL,
    ->     vehicle_id   INT,
    ->     assigned_qty INT NOT NULL CHECK (assigned_qty > 0),
    ->     eta_minutes  INT,
    ->     matched_at   TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     FOREIGN KEY (request_id) REFERENCES request(request_id),
    ->     FOREIGN KEY (supply_id) REFERENCES supply(supply_id),
    ->     FOREIGN KEY (vehicle_id) REFERENCES vehicle(vehicle_id)
    -> );
Query OK, 0 rows affected (0.05 sec)
```

```
(PES2UG24CS655@root@localhost) [disaster] >  CREATE TABLE request_history (
    ->     history_id   INT PRIMARY KEY AUTO_INCREMENT,
    ->     request_id   INT NOT NULL,
    ->     old_status   VARCHAR(20),
    ->     new_status   VARCHAR(20),
    ->     changed_at   TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     FOREIGN KEY (request_id) REFERENCES request(request_id)
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE users (
    ->     user_id        INT PRIMARY KEY AUTO_INCREMENT,
    ->     username       VARCHAR(50) NOT NULL UNIQUE,
    ->     full_name      VARCHAR(150),
    ->     password_hash  VARCHAR(255) NOT NULL,
    ->     role           ENUM('admin','logistics_manager','field_worker') NOT NULL DEFAULT 'field_worker',
    ->     created_at     TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     CHECK (CHAR_LENGTH(username) >= 3),
    ->     CHECK (CHAR_LENGTH(password_hash) = 60 AND password_hash LIKE '$2_$%')
    -> );
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> desc users;
+---------------+-----------------------------------------------+------+-----+-------------------+-------------------+
| Field         | Type                                          | Null | Key | Default           | Extra             |
+---------------+-----------------------------------------------+------+-----+-------------------+-------------------+
| user_id       | int                                           | NO   | PRI | NULL              | auto_increment    |
| username      | varchar(50)                                   | NO   | UNI | NULL              |                   |
| full_name     | varchar(150)                                  | YES  |     | NULL              |                   |
| password_hash | varchar(255)                                  | NO   |     | NULL              |                   |
| role          | enum('admin','logistics_manager','field_worker') | NO   |     | field_worker      |                   |
| created_at    | timestamp                                     | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+---------------+-----------------------------------------------+------+-----+-------------------+-------------------+
6 rows in set (0.00 sec)
```

```
mysql> show tables;
+--------------------+
| Tables_in_disaster |
+--------------------+
| depot              |
| match_assignment   |
| request            |
| request_history    |
| route              |
| shelter            |
| supply             |
| users              |
| vehicle            |
| victim             |
+--------------------+
10 rows in set (0.00 sec)
```

```
(PES2UG24CS655@root@localhost) [disaster] >  desc depot;
+-----------+--------------+------+-----+---------+----------------+
| Field     | Type         | Null | Key | Default | Extra          |
+-----------+--------------+------+-----+---------+----------------+
| depot_id  | int          | NO   | PRI | NULL    | auto_increment |
| name      | varchar(150) | NO   |     | NULL    |                |
| latitude  | decimal(9,6) | NO   |     | NULL    |                |
| longitude | decimal(9,6) | NO   |     | NULL    |                |
+-----------+--------------+------+-----+---------+----------------+
4 rows in set (0.01 sec)

(PES2UG24CS655@root@localhost) [disaster] >  desc match_assignment;
+--------------+-----------+------+-----+-------------------+-------------------+
| Field        | Type      | Null | Key | Default           | Extra             |
+--------------+-----------+------+-----+-------------------+-------------------+
| match_id     | int       | NO   | PRI | NULL              | auto_increment    |
| request_id   | int       | NO   | UNI | NULL              |                   |
| supply_id    | int       | NO   | MUL | NULL              |                   |
| vehicle_id   | int       | YES  | MUL | NULL              |                   |
| assigned_qty | int       | NO   |     | NULL              |                   |
| eta_minutes  | int       | YES  |     | NULL              |                   |
| matched_at   | timestamp | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+--------------+-----------+------+-----+-------------------+-------------------+
7 rows in set (0.01 sec)

(PES2UG24CS655@root@localhost) [disaster] >  desc request;
+---------------+-----------------------------------------------+------+-----+-------------------+-------------------+
| Field         | Type                                          | Null | Key | Default           | Extra             |
+---------------+-----------------------------------------------+------+-----+-------------------+-------------------+
| request_id    | int                                           | NO   | PRI | NULL              | auto_increment    |
| person_id     | int                                           | NO   | MUL | NULL              |                   |
| resource_type | varchar(50)                                   | NO   |     | NULL              |                   |
| quantity      | int                                           | NO   |     | NULL              |                   |
| status        | enum('pending','matched','delivered','cancelled') | YES  |     | pending           |                   |
| requested_at  | timestamp                                     | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| description   | text                                          | YES  |     | NULL              |                   |
+---------------+-----------------------------------------------+------+-----+-------------------+-------------------+
7 rows in set (0.00 sec)
```

```
(PES2UG24CS655@root@localhost) [disaster] >  desc request_history;
+------------+-------------+------+-----+-------------------+-------------------+
| Field      | Type        | Null | Key | Default           | Extra             |
+------------+-------------+------+-----+-------------------+-------------------+
| history_id | int         | NO   | PRI | NULL              | auto_increment    |
| request_id | int         | NO   | MUL | NULL              |                   |
| old_status | varchar(20) | YES  |     | NULL              |                   |
| new_status | varchar(20) | YES  |     | NULL              |                   |
| changed_at | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------+-------------+------+-----+-------------------+-------------------+
5 rows in set (0.01 sec)

(PES2UG24CS655@root@localhost) [disaster] >  desc route;
+-------------+-----------------------+------+-----+---------+----------------+
| Field       | Type                  | Null | Key | Default | Extra          |
+-------------+-----------------------+------+-----+---------+----------------+
| route_id    | int                   | NO   | PRI | NULL    | auto_increment |
| depot_id    | int                   | NO   | MUL | NULL    |                |
| shelter_id  | int                   | NO   | MUL | NULL    |                |
| distance_km | decimal(6,2)          | YES  |     | NULL    |                |
| status      | enum('open','blocked') | YES  |     | open    |                |
+-------------+-----------------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)

(PES2UG24CS655@root@localhost) [disaster] >  desc shelter;
+------------+--------------+------+-----+---------+----------------+
| Field      | Type         | Null | Key | Default | Extra          |
+------------+--------------+------+-----+---------+----------------+
| shelter_id | int          | NO   | PRI | NULL    | auto_increment |
| name       | varchar(150) | NO   |     | NULL    |                |
| latitude   | decimal(9,6) | NO   |     | NULL    |                |
| longitude  | decimal(9,6) | NO   |     | NULL    |                |
| capacity   | int          | NO   |     | NULL    |                |
| occupancy  | int          | YES  |     | 0       |                |
+------------+--------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)
```

```
(PES2UG24CS655@root@localhost) [disaster] >  desc supply;
+---------------+-------------+------+-----+---------+----------------+
| Field         | Type        | Null | Key | Default | Extra          |
+---------------+-------------+------+-----+---------+----------------+
| supply_id     | int         | NO   | PRI | NULL    | auto_increment |
| depot_id      | int         | YES  | MUL | NULL    |                |
| shelter_id    | int         | YES  | MUL | NULL    |                |
| resource_type | varchar(50) | NO   |     | NULL    |                |
| quantity      | int         | NO   |     | NULL    |                |
+---------------+-------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)

(PES2UG24CS655@root@localhost) [disaster] >  desc vehicle;
+------------+-----------------------------------------+------+-----+-----------+----------------+
| Field      | Type                                    | Null | Key | Default   | Extra          |
+------------+-----------------------------------------+------+-----+-----------+----------------+
| vehicle_id | int                                     | NO   | PRI | NULL      | auto_increment |
| status     | enum('available','en_route','maintenance') | YES  |     | available |                |
| latitude   | decimal(9,6)                            | YES  |     | NULL      |                |
| longitude  | decimal(9,6)                            | YES  |     | NULL      |                |
| capacity   | int                                     | NO   |     | NULL      |                |
+------------+-----------------------------------------+------+-----+-----------+----------------+
5 rows in set (0.01 sec)

(PES2UG24CS655@root@localhost) [disaster] >  desc victim;
+------------+--------------------------------------+------+-----+---------+----------------+
| Field      | Type                                 | Null | Key | Default | Extra          |
+------------+--------------------------------------+------+-----+---------+----------------+
| person_id  | int                                  | NO   | PRI | NULL    | auto_increment |
| name       | varchar(150)                         | YES  |     | NULL    |                |
| latitude   | decimal(9,6)                         | NO   |     | NULL    |                |
| longitude  | decimal(9,6)                         | NO   |     | NULL    |                |
| group_size | int                                  | YES  |     | 1       |                |
| priority   | enum('critical','high','medium','low') | YES  |     | medium  |                |
+------------+--------------------------------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)
```

```
DROP DATABASE IF EXISTS disaster;
CREATE DATABASE disaster;
USE disaster;

CREATE TABLE victim (
   person_id    INT PRIMARY KEY AUTO_INCREMENT,
   name         VARCHAR(150),
   latitude     DECIMAL(9,6) NOT NULL,
   longitude    DECIMAL(9,6) NOT NULL,
   group_size   INT DEFAULT 1,
   priority     ENUM('critical','high','medium','low') DEFAULT 'medium'
);

CREATE TABLE depot (
   depot_id     INT PRIMARY KEY AUTO_INCREMENT,
   name         VARCHAR(150) NOT NULL,
   latitude     DECIMAL(9,6) NOT NULL,
   longitude    DECIMAL(9,6) NOT NULL
);

CREATE TABLE shelter (
   shelter_id   INT PRIMARY KEY AUTO_INCREMENT,
   name         VARCHAR(150) NOT NULL,
   latitude     DECIMAL(9,6) NOT NULL,
   longitude    DECIMAL(9,6) NOT NULL,
   capacity     INT NOT NULL,
   occupancy    INT DEFAULT 0,
   CHECK (occupancy >= 0 AND occupancy <= capacity)
);

CREATE TABLE route (
   route_id     INT PRIMARY KEY AUTO_INCREMENT,
   depot_id     INT NOT NULL,
   shelter_id   INT NOT NULL,
   distance_km  DECIMAL(6,2),
   status       ENUM('open','blocked') DEFAULT 'open',
   FOREIGN KEY (depot_id) REFERENCES depot(depot_id),
   FOREIGN KEY (shelter_id) REFERENCES shelter(shelter_id)
);

CREATE TABLE supply (
   supply_id    INT PRIMARY KEY AUTO_INCREMENT,
   depot_id     INT,
   shelter_id   INT,
   resource_type VARCHAR(50) NOT NULL,
   quantity     INT NOT NULL,
   CHECK (quantity >= 0),
   CHECK ((shelter_id IS NOT NULL AND depot_id IS NULL)
      OR (shelter_id IS NULL AND depot_id IS NOT NULL)),
```

```
    UNIQUE (depot_id, resource_type),
    UNIQUE (shelter_id, resource_type),
    FOREIGN KEY (depot_id) REFERENCES depot(depot_id),
    FOREIGN KEY (shelter_id) REFERENCES shelter(shelter_id)
);

CREATE TABLE vehicle (
    vehicle_id   INT PRIMARY KEY AUTO_INCREMENT,
    status       ENUM('available','en_route','maintenance') DEFAULT 'available',
    latitude     DECIMAL(9,6),
    longitude    DECIMAL(9,6),
    capacity     INT NOT NULL
);

CREATE TABLE request (
    request_id    INT PRIMARY KEY AUTO_INCREMENT,
    person_id     INT NOT NULL,
    resource_type VARCHAR(50) NOT NULL,
    quantity      INT NOT NULL CHECK (quantity > 0),
    status        ENUM('pending','matched','delivered','cancelled') DEFAULT 'pending',
    requested_at  TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    description   TEXT,
    FOREIGN KEY (person_id) REFERENCES victim(person_id)
);

CREATE TABLE match_assignment (
    match_id     INT PRIMARY KEY AUTO_INCREMENT,
    request_id   INT NOT NULL UNIQUE,
    supply_id    INT NOT NULL,
    vehicle_id   INT,
    assigned_qty INT NOT NULL CHECK (assigned_qty > 0),
    eta_minutes  INT,
    matched_at   TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (request_id) REFERENCES request(request_id),
    FOREIGN KEY (supply_id)  REFERENCES supply(supply_id),
    FOREIGN KEY (vehicle_id) REFERENCES vehicle(vehicle_id)
);

CREATE TABLE request_history (
    history_id   INT PRIMARY KEY AUTO_INCREMENT,
    request_id   INT NOT NULL,
    old_status   VARCHAR(20),
    new_status   VARCHAR(20),
    changed_at   TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (request_id) REFERENCES request(request_id)
);

CREATE TABLE users (
    user_id       INT PRIMARY KEY AUTO_INCREMENT,
```

```
    username       VARCHAR(50) NOT NULL UNIQUE,
    full_name      VARCHAR(150),
    password_hash  VARCHAR(255) NOT NULL,
    role           ENUM('admin','logistics_manager','field_worker') NOT NULL DEFAULT 'field_worker',
    created_at     TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    CHECK (CHAR_LENGTH(username) >= 3),
    CHECK (CHAR_LENGTH(password_hash) = 60 AND password_hash LIKE '$2_$%')
);
```

6. DML STATEMENTS

STATEMENTS WITH SCREEN SHOTS OF THE TABLE WITH INSERTED VALUES

```
mysql> select * from depot;
+----------+------------------------+----------+-----------+
| depot_id | name                   | latitude | longitude |
+----------+------------------------+----------+-----------+
|        1 | Central Depot          | 12.960000 | 77.600000 |
|        2 | North Depot            | 13.030000 | 77.590000 |
|        3 | West Depot             | 12.955000 | 77.570000 |
|        4 | South East Depot       | 12.900000 | 77.650000 |
|        5 | Hill View Depot        | 13.040000 | 77.620000 |
|        6 | Lake Side Depot        | 12.880000 | 77.610000 |
|        7 | Industrial Depot       | 12.920000 | 77.670000 |
|        8 | Metro Depot            | 12.985000 | 77.640000 |
|        9 | Airport Relief Depot   | 13.100000 | 77.590000 |
|       10 | Railway Relief Depot   | 12.975000 | 77.580000 |
|       11 | Market Depot           | 12.950000 | 77.690000 |
|       12 | University Depot       | 12.940000 | 77.560000 |
|       13 | Community Depot        | 13.020000 | 77.650000 |
|       14 | Emergency Depot        | 12.910000 | 77.570000 |
|       15 | East Relief Depot      | 12.970000 | 77.700000 |
|       16 | South Relief Depot     | 12.860000 | 77.640000 |
|       17 | North Relief Depot     | 13.060000 | 77.570000 |
|       18 | Central Emergency Depot | 12.965000 | 77.590000 |
|       19 | Ring Road Depot        | 12.950000 | 77.720000 |
|       20 | Garden City Depot      | 12.915000 | 77.625000 |
+----------+------------------------+----------+-----------+
20 rows in set (0.00 sec)

mysql> select * from request;
+------------+-----------+---------------+----------+-----------+---------------------+----------------------------------------+
| request_id | person_id | resource_type | quantity | status    | requested_at        | description                            |
+------------+-----------+---------------+----------+-----------+---------------------+----------------------------------------+
|          1 |         1 | water         |       40 | pending   | 2026-11-02 17:19:35 | Flooded area, no drinking water        |
|          2 |         2 | medicine      |       10 | pending   | 2026-11-02 17:19:35 | Injured family                         |
|          3 |         3 | food          |       20 | pending   | 2026-11-02 17:19:35 | Group of 6 stranded                    |
|          4 |         4 | water         |       30 | pending   | 2026-11-02 17:19:35 | Low urgency                            |
|          5 |        11 | water         |       20 | pending   | 2026-11-02 17:19:35 | Drinking water required for family     |
|          6 |        12 | food          |       30 | pending   | 2026-11-02 17:19:35 | Food packets required urgently         |
|          7 |        13 | medicine      |       10 | pending   | 2026-11-02 17:19:35 | Basic medical supplies needed          |
|          8 |        14 | blankets      |       15 | pending   | 2026-11-02 17:19:35 | Blankets required for shelter          |
|          9 |        15 | water         |       25 | pending   | 2026-11-02 17:19:35 | Water shortage in affected area        |
|         10 |        16 | food          |       40 | pending   | 2026-11-02 17:19:35 | Food required for large family         |
|         11 |        17 | medicine      |       20 | pending   | 2026-11-02 17:19:35 | Medicine required for injured residents |
|         12 |        18 | water         |       35 | pending   | 2026-11-02 17:19:35 | Emergency drinking water request       |
|         13 |        19 | blankets      |       10 | pending   | 2026-11-02 17:19:35 | Blankets required for children         |
|         14 |        20 | food          |       15 | pending   | 2026-11-02 17:19:35 | Food required for stranded group       |
|         15 |        21 | water         |       50 | pending   | 2026-11-02 17:19:35 | Large group needs drinking water       |
|         16 |        22 | medicine      |       25 | pending   | 2026-11-02 17:19:35 | Medical supplies urgently required     |
|         17 |        23 | food          |       35 | pending   | 2026-11-02 17:19:35 | Food packets needed                    |
|         18 |        24 | blankets      |       30 | pending   | 2026-11-02 17:19:35 | Blankets required at relief camp       |
|         19 |        25 | water         |       45 | pending   | 2026-11-02 17:19:35 | Drinking water required                |
|         20 |        11 | food          |       20 | pending   | 2026-11-02 17:19:35 | Additional food requirement            |
|         21 |        13 | water         |       15 | pending   | 2026-11-02 17:19:35 | Water required for household           |
|         22 |        15 | medicine      |        8 | pending   | 2026-11-02 17:19:35 | First aid supplies required            |
|         23 |        18 | food          |       25 | pending   | 2026-11-02 17:19:35 | Food needed for displaced family       |
|         24 |        22 | blankets      |       12 | pending   | 2026-11-02 17:19:35 | Blankets needed at temporary shelter   |
|         26 |         1 | smoke-item    |        8 | cancelled | 2026-11-02 17:29:11 | SMOKE TEST request                     |
|         28 |         1 | smoke-item    |        8 | cancelled | 2026-11-02 17:58:44 | SMOKE TEST request                     |
+------------+-----------+---------------+----------+-----------+---------------------+----------------------------------------+
```

```
+------------+----------+--------------+----------+----------+--------------------+--------------------------------------+
26 rows in set (0.00 sec)

mysql> select * from route;
+----------+----------+------------+-------------+---------+
| route_id | depot_id | shelter_id | distance_km | status  |
+----------+----------+------------+-------------+---------+
|        1 |        1 |          1 |        3.50 | open    |
|        2 |        1 |          2 |        5.20 | open    |
|        3 |        2 |          2 |        7.80 | blocked |
|        4 |        5 |          6 |        4.20 | open    |
|        5 |        6 |          7 |        6.80 | open    |
|        6 |        7 |          8 |        9.50 | blocked |
|        7 |        8 |          9 |        5.40 | open    |
|        8 |        9 |         10 |        7.20 | open    |
|        9 |       10 |         11 |        3.80 | open    |
|       10 |       11 |         12 |        8.10 | blocked |
|       11 |       12 |         13 |        6.50 | open    |
|       12 |       13 |         14 |        5.90 | open    |
|       13 |       14 |         15 |        4.70 | open    |
|       14 |       15 |         16 |       10.20 | blocked |
|       15 |       16 |         17 |        7.60 | open    |
|       16 |       17 |         18 |        5.10 | open    |
|       17 |       18 |         19 |        8.40 | open    |
|       18 |       19 |         20 |        6.30 | blocked |
|       19 |       20 |          1 |        4.90 | open    |
|       20 |        3 |          6 |        5.50 | open    |
|       21 |        4 |         10 |        7.90 | blocked |
|       22 |        6 |         15 |        9.10 | open    |
|       23 |        8 |         20 |        6.60 | open    |
+----------+----------+------------+-------------+---------+
23 rows in set (0.00 sec)

mysql> select * from shelter;
+------------+--------------------------+-----------+-----------+----------+-----------+
| shelter_id | name                     | latitude  | longitude | capacity | occupancy |
+------------+--------------------------+-----------+-----------+----------+-----------+
|          1 | City School Shelter      | 12.940000 | 77.610000 |      200 |        50 |
|          2 | Stadium Shelter          | 12.985000 | 77.570000 |      500 |       120 |
|          3 | Temple Shelter           | 12.960000 | 77.580000 |      250 |        60 |
|          4 | Government School Shelter | 12.930000 | 77.620000 |      350 |       120 |
|          5 | Community Centre Shelter | 12.980000 | 77.660000 |      180 |        70 |
|          6 | Public Hall Shelter      | 12.900000 | 77.600000 |      220 |        90 |
|          7 | College Auditorium       | 12.950000 | 77.670000 |      500 |       250 |
|          8 | Municipal Shelter        | 13.020000 | 77.610000 |      300 |       100 |
|          9 | Sports Stadium North     | 13.050000 | 77.600000 |      600 |       300 |
|         10 | Village Community Hall   | 12.870000 | 77.620000 |      150 |        50 |
|         11 | Metro Community Shelter  | 12.990000 | 77.650000 |      275 |        80 |
|         12 | Relief Camp Alpha        | 12.910000 | 77.660000 |      400 |       150 |
|         13 | Relief Camp Beta         | 12.940000 | 77.690000 |      450 |       200 |
|         14 | Relief Camp Gamma        | 13.030000 | 77.680000 |      350 |        90 |
|         15 | Relief Camp Delta        | 12.880000 | 77.570000 |      200 |        75 |
|         16 | District Shelter         | 13.070000 | 77.590000 |      500 |       220 |
|         17 | Emergency Shelter        | 12.920000 | 77.550000 |      175 |        40 |
|         18 | Ward Community Shelter   | 13.025000 | 77.575000 |      300 |       110 |
|         19 | Eastside Relief Camp     | 12.965000 | 77.650000 |      350 |       130 |
|         20 | Lakeside Hall Shelter    | 12.990000 | 77.690000 |      250 |        90 |
+------------+--------------------------+-----------+-----------+----------+-----------+
20 rows in set (0.00 sec)
```

```
+------------+--------------------------+-----------+-----------+----------+----------+
20 rows in set (0.00 sec)

mysql> select * from supply;
+-----------+----------+------------+--------------+----------+
| supply_id | depot_id | shelter_id | resource_type | quantity |
+-----------+----------+------------+--------------+----------+
|         1 |        1 |       NULL | water         |      100 |
|         2 |        2 |       NULL | water         |       60 |
|         3 |     NULL |          1 | medicine      |       30 |
|         4 |     NULL |          2 | blankets      |       40 |
|         5 |        5 |       NULL | water         |      150 |
|         6 |        6 |       NULL | food          |      100 |
|         7 |        7 |       NULL | medicine      |       75 |
|         8 |        8 |       NULL | blankets      |       90 |
|         9 |        9 |       NULL | water         |      200 |
|        10 |       10 |       NULL | food          |      120 |
|        11 |       11 |       NULL | medicine      |       60 |
|        12 |       12 |       NULL | blankets      |      110 |
|        13 |       13 |       NULL | water         |      180 |
|        14 |       14 |       NULL | food          |       90 |
|        15 |       15 |       NULL | medicine      |       80 |
|        16 |       16 |       NULL | blankets      |       70 |
|        17 |     NULL |          6 | water         |       50 |
|        18 |     NULL |          7 | food          |       75 |
|        19 |     NULL |          8 | medicine      |       40 |
|        20 |     NULL |          9 | blankets      |       60 |
|        21 |     NULL |         10 | water         |      100 |
|        22 |     NULL |         11 | food          |       55 |
|        23 |     NULL |         12 | medicine      |       35 |
|        24 |     NULL |         13 | blankets      |       80 |
+-----------+----------+------------+--------------+----------+
24 rows in set (0.00 sec)

mysql> select * from users;
+---------+-------------+---------------------+--------------------------------------------------------------+------------------+---------------------+
| user_id | username    | full_name           | password_hash                                                | role             | created_at          |
+---------+-------------+---------------------+--------------------------------------------------------------+------------------+---------------------+
|       1 | admin       | System Administrator | $2b$10$ZgHzW50/4sEyTDN/WtRvOO7mMv5rqbDQQBr2Jxj4TAoqgl/HTj9K. | admin            | 2026-11-02 17:19:35 |
|       2 | logistics   | Logistics Manager    | $2b$10$zacEYkVoSbjUNt8lxqOk9.AqJjw3YxKHbwy5CjruzvsJxJBCzZOfS | logistics_manager | 2026-11-02 17:19:35 |
|       3 | fieldworker | Field Worker         | $2b$10$33Y3T.9H9pUj7ny6RTfrcOXqHEu/2XAh2Tm/j.5hdk1tEt0qRZ7X. | field_worker     | 2026-11-02 17:19:35 |
+---------+-------------+---------------------+--------------------------------------------------------------+------------------+---------------------+
3 rows in set (0.00 sec)

mysql> select * from vehicle;
+------------+-------------+-----------+-----------+----------+
| vehicle_id | status      | latitude  | longitude | capacity |
+------------+-------------+-----------+-----------+----------+
|          1 | available   | 12.962000 | 77.602000 |      100 |
|          2 | available   | 13.000000 | 77.580000 |       50 |
|          3 | maintenance | 12.950000 | 77.600000 |      200 |
|          4 | available   | 12.960000 | 77.610000 |      100 |
|          5 | available   | 12.950000 | 77.620000 |       75 |
|          6 | available   | 12.940000 | 77.630000 |      150 |
|          7 | available   | 12.930000 | 77.640000 |       50 |
|          8 | available   | 12.920000 | 77.650000 |      200 |
|          9 | available   | 12.980000 | 77.620000 |      120 |
|         10 | available   | 13.000000 | 77.630000 |       90 |
|         11 | available   | 13.020000 | 77.640000 |      180 |
|         12 | available   | 13.040000 | 77.650000 |      250 |
|         13 | available   | 13.060000 | 77.600000 |      100 |
|         14 | maintenance | 12.900000 | 77.600000 |      150 |
|         15 | available   | 12.880000 | 77.620000 |       80 |
|         16 | available   | 12.910000 | 77.660000 |      130 |
|         17 | available   | 12.930000 | 77.680000 |       70 |
|         18 | available   | 12.950000 | 77.700000 |      220 |
|         19 | maintenance | 12.970000 | 77.680000 |      200 |
|         20 | available   | 13.010000 | 77.670000 |      160 |
|         21 | available   | 13.030000 | 77.690000 |      100 |
|         22 | available   | 12.890000 | 77.650000 |       60 |
|         23 | available   | 12.870000 | 77.630000 |      140 |
+------------+-------------+-----------+-----------+----------+
```

```
mysql> select * from request_history;
+------------+------------+------------+------------+---------------------+
| history_id | request_id | old_status | new_status | changed_at          |
+------------+------------+------------+------------+---------------------+
|          1 |         26 | NULL       | pending    | 2026-11-02 17:29:11 |
|          2 |         26 | pending    | cancelled  | 2026-11-02 17:29:11 |
|          3 |         28 | NULL       | pending    | 2026-11-02 17:58:44 |
|          4 |         28 | pending    | cancelled  | 2026-11-02 17:58:44 |
+------------+------------+------------+------------+---------------------+
4 rows in set (0.00 sec)

mysql> S
```

```
mysql> select * from victim;
+-----------+---------+-----------+-----------+------------+----------+
| person_id | name    | latitude  | longitude | group_size | priority |
+-----------+---------+-----------+-----------+------------+----------+
|         1 | Asha    | 12.971600 | 77.594600 |          4 | critical |
|         2 | Ravi    | 12.935200 | 77.624500 |          2 | high     |
|         3 | Meena   | 12.958000 | 77.640000 |          6 | medium   |
|         4 | Karan   | 12.990000 | 77.550000 |          1 | low      |
|         5 | Neha    | 12.965000 | 77.605000 |          3 | medium   |
|         6 | Vikram  | 12.975000 | 77.615000 |          8 | critical |
|         7 | Pooja   | 12.930000 | 77.590000 |          2 | low      |
|         8 | Manoj   | 12.985000 | 77.650000 |          5 | high     |
|         9 | Sneha   | 12.945000 | 77.625000 |          4 | medium   |
|        10 | Ramesh  | 12.905000 | 77.610000 |          6 | critical |
|        11 | Kavya   | 13.015000 | 77.585000 |          2 | high     |
|        12 | Deepak  | 12.955000 | 77.675000 |          3 | low      |
|        13 | Nisha   | 12.925000 | 77.645000 |          5 | medium   |
|        14 | Amit    | 12.995000 | 77.600000 |          7 | critical |
|        15 | Swati   | 12.935000 | 77.655000 |          2 | high     |
|        16 | Harish  | 12.885000 | 77.590000 |          4 | medium   |
|        17 | Preeti  | 13.025000 | 77.630000 |          3 | low      |
|        18 | Sanjay  | 12.970000 | 77.680000 |          6 | high     |
|        19 | Aarti   | 12.915000 | 77.650000 |          2 | medium   |
|        20 | Tara    | 12.900000 | 77.625000 |          6 | high     |
|        21 | Imran   | 12.960000 | 77.665000 |         10 | critical |
|        22 | Lakshmi | 13.005000 | 77.570000 |          7 | critical |
|        23 | Joseph  | 12.870000 | 77.600000 |          4 | medium   |
|        24 | Zoya    | 12.945000 | 77.705000 |          8 | medium   |
|        25 | Girish  | 13.040000 | 77.640000 |          5 | high     |
+-----------+---------+-----------+-----------+------------+----------+
25 rows in set (0.00 sec)
```

-- ==
-- 1. VICTIM
-- Final: 25
-- ==

INSERT INTO victim
(name, latitude, longitude, group_size, priority)
VALUES
('Asha',12.971600,77.594600,4,'critical'),
('Ravi',12.935200,77.624500,2,'high'),
('Meena',12.958000,77.640000,6,'medium'),
('Karan',12.990000,77.550000,1,'low'),
('Neha',12.965000,77.605000,3,'medium'),
('Vikram',12.975000,77.615000,8,'critical'),
('Pooja',12.930000,77.590000,2,'low'),
('Manoj',12.985000,77.650000,5,'high'),
('Sneha',12.945000,77.625000,4,'medium'),
('Ramesh',12.905000,77.610000,6,'critical'),
('Kavya',13.015000,77.585000,2,'high'),
('Deepak',12.955000,77.675000,3,'low'),
('Nisha',12.925000,77.645000,5,'medium'),
('Amit',12.995000,77.600000,7,'critical'),
('Swati',12.935000,77.655000,2,'high'),
('Harish',12.885000,77.590000,4,'medium'),
('Preeti',13.025000,77.630000,3,'low'),
('Sanjay',12.970000,77.680000,6,'high'),
('Aarti',12.915000,77.650000,2,'medium'),
('Tara',12.900000,77.625000,6,'high'),
('Imran',12.960000,77.665000,10,'critical'),

('Lakshmi',13.005000,77.570000,7,'critical'),
('Joseph',12.870000,77.600000,4,'medium'),
('Zoya',12.945000,77.705000,8,'medium'),
('Girish',13.040000,77.640000,5,'high');

-- ==
-- 2. DEPOT
-- Final: 20
-- ==

INSERT INTO depot
(name, latitude, longitude)
VALUES
('Central Depot',12.960000,77.600000),
('North Depot',13.030000,77.590000),
('West Depot',12.955000,77.570000),
('South East Depot',12.900000,77.650000),
('Hill View Depot',13.040000,77.620000),
('Lake Side Depot',12.880000,77.610000),
('Industrial Depot',12.920000,77.670000),
('Metro Depot',12.985000,77.640000),
('Airport Relief Depot',13.100000,77.590000),
('Railway Relief Depot',12.975000,77.580000),
('Market Depot',12.950000,77.690000),
('University Depot',12.940000,77.560000),
('Community Depot',13.020000,77.650000),
('Emergency Depot',12.910000,77.570000),
('East Relief Depot',12.970000,77.700000),
('South Relief Depot',12.860000,77.640000),
('North Relief Depot',13.060000,77.570000),
('Central Emergency Depot',12.965000,77.590000),
('Ring Road Depot',12.950000,77.720000),
('Garden City Depot',12.915000,77.625000);

-- ==
-- 3. SHELTER
-- Final: 20
-- ==

INSERT INTO shelter
(name, latitude, longitude, capacity, occupancy)
VALUES
('City School Shelter',12.940000,77.610000,200,50),
('Stadium Shelter',12.985000,77.570000,500,120),
('Temple Shelter',12.960000,77.580000,250,60),
('Government School Shelter',12.930000,77.620000,350,120),
('Community Centre Shelter',12.980000,77.660000,180,70),

('Public Hall Shelter',12.900000,77.600000,220,90),
('College Auditorium',12.950000,77.670000,500,250),
('Municipal Shelter',13.020000,77.610000,300,100),
('Sports Stadium North',13.050000,77.600000,600,300),
('Village Community Hall',12.870000,77.620000,150,50),
('Metro Community Shelter',12.990000,77.650000,275,80),
('Relief Camp Alpha',12.910000,77.660000,400,150),
('Relief Camp Beta',12.940000,77.690000,450,200),
('Relief Camp Gamma',13.030000,77.680000,350,90),
('Relief Camp Delta',12.880000,77.570000,200,75),
('District Shelter',13.070000,77.590000,500,220),
('Emergency Shelter',12.920000,77.550000,175,40),
('Ward Community Shelter',13.025000,77.575000,300,110),
('Eastside Relief Camp',12.965000,77.650000,350,130),
('Lakeside Hall Shelter',12.990000,77.690000,250,90);

-- ==
-- 4. ROUTE
-- Final: 23
-- ==

INSERT INTO route
(depot_id, shelter_id, distance_km, status)
VALUES
(1,1,3.5,'open'),
(1,2,5.2,'open'),
(2,2,7.8,'blocked'),
(5,6,4.2,'open'),
(6,7,6.8,'open'),
(7,8,9.5,'blocked'),
(8,9,5.4,'open'),
(9,10,7.2,'open'),
(10,11,3.8,'open'),
(11,12,8.1,'blocked'),
(12,13,6.5,'open'),
(13,14,5.9,'open'),
(14,15,4.7,'open'),
(15,16,10.2,'blocked'),
(16,17,7.6,'open'),
(17,18,5.1,'open'),
(18,19,8.4,'open'),
(19,20,6.3,'blocked'),
(20,1,4.9,'open'),
(3,6,5.5,'open'),
(4,10,7.9,'blocked'),
(6,15,9.1,'open'),
(8,20,6.6,'open');

```
-- ============================================================
-- 5. SUPPLY
-- Final: 24
-- ============================================================

INSERT INTO supply
(depot_id, shelter_id, resource_type, quantity)
VALUES
(1,NULL,'water',100),
(2,NULL,'water',60),
(NULL,1,'medicine',30),
(NULL,2,'blankets',40),
(5,NULL,'water',150),
(6,NULL,'food',100),
(7,NULL,'medicine',75),
(8,NULL,'blankets',90),
(9,NULL,'water',200),
(10,NULL,'food',120),
(11,NULL,'medicine',60),
(12,NULL,'blankets',110),
(13,NULL,'water',180),
(14,NULL,'food',90),
(15,NULL,'medicine',80),
(16,NULL,'blankets',70),
(NULL,6,'water',50),
(NULL,7,'food',75),
(NULL,8,'medicine',40),
(NULL,9,'blankets',60),
(NULL,10,'water',100),
(NULL,11,'food',55),
(NULL,12,'medicine',35),
(NULL,13,'blankets',80);


-- ============================================================
-- 6. VEHICLE
-- Final: 23
-- ============================================================

INSERT INTO vehicle
(status, latitude, longitude, capacity)
VALUES
('available',12.962000,77.602000,100),
('available',13.000000,77.580000,50),
('maintenance',12.950000,77.600000,200),
('available',12.960000,77.610000,100),
('available',12.950000,77.620000,75),
('available',12.940000,77.630000,150),
```

('available',12.930000,77.640000,50),
('available',12.920000,77.650000,200),
('available',12.980000,77.620000,120),
('available',13.000000,77.630000,90),
('available',13.020000,77.640000,180),
('available',13.040000,77.650000,250),
('available',13.060000,77.600000,100),
('maintenance',12.900000,77.600000,150),
('available',12.880000,77.620000,80),
('available',12.910000,77.660000,130),
('available',12.930000,77.680000,70),
('available',12.950000,77.700000,220),
('maintenance',12.970000,77.680000,200),
('available',13.010000,77.670000,160),
('available',13.030000,77.690000,100),
('available',12.890000,77.650000,60),
('available',12.870000,77.630000,140);

-- ==
-- 7. REQUEST
-- Final: 24
--
-- request_history entries are automatically created
-- by trg_req_ins for every request inserted.
-- ==

INSERT INTO request
(person_id, resource_type, quantity, description)
VALUES
(1,'water',40,'Flooded area, no drinking water'),
(2,'medicine',10,'Injured family'),
(3,'food',20,'Group of 6 stranded'),
(4,'water',30,'Low urgency'),
(11,'water',20,'Drinking water required for family'),
(12,'food',30,'Food packets required urgently'),
(13,'medicine',10,'Basic medical supplies needed'),
(14,'blankets',15,'Blankets required for shelter'),
(15,'water',25,'Water shortage in affected area'),
(16,'food',40,'Food required for large family'),
(17,'medicine',20,'Medicine required for injured residents'),
(18,'water',35,'Emergency drinking water request'),
(19,'blankets',10,'Blankets required for children'),
(20,'food',15,'Food required for stranded group'),
(21,'water',50,'Large group needs drinking water'),
(22,'medicine',25,'Medical supplies urgently required'),
(23,'food',35,'Food packets needed'),
(24,'blankets',30,'Blankets required at relief camp'),
(25,'water',45,'Drinking water required'),

```
(11,'food',20,'Additional food requirement'),
(13,'water',15,'Water required for household'),
(15,'medicine',8,'First aid supplies required'),
(18,'food',25,'Food needed for displaced family'),
(22,'blankets',12,'Blankets needed at temporary shelter');


-- ============================================================
-- 8. USERS
-- Final: 3 (one per role)
--
-- password_hash holds bcrypt hashes only (enforced by CHECK).
-- DEV/DEMO seed accounts: change these passwords before any
-- real deployment.
-- ============================================================

INSERT INTO users
(username, full_name, password_hash, role)
VALUES
('admin','System Administrator','2b
10$ZgHzW50/4sEyTDN/WtRvOO7mMv5rqbDQQBr2Jxj4TAoqgl/HTj9K.','admin'),
('logistics','Logistics Manager','2b
10$zacEYkVoSbjUNt8lxqOk9.AqJjw3YxKHbwy5CjruzvsJxJBCzZOfS','logistics_manager'),
('fieldworker','Field Worker','2b
10$33Y3T.9H9pUj7ny6RTfrcOXqHEu/2XAh2Tm/j.5hdk1tEt0qRZ7X.','field_worker');
```

7. QUERIES

7.1 SIMPLE QUERY WITH GROUP BY, AGGREGATE

```
mysql> SELECT * FROM route LIMIT 5;
+----------+----------+------------+-------------+---------+
| route_id | depot_id | shelter_id | distance_km | status  |
+----------+----------+------------+-------------+---------+
|        1 |        1 |          1 |        3.50 | open    |
|        2 |        1 |          2 |        5.20 | open    |
|        3 |        2 |          2 |        7.80 | blocked |
|       17 |        2 |          3 |        8.00 | open    |
+----------+----------+------------+-------------+---------+
4 rows in set (0.000 sec)

mysql> SELECT
    ->     status,
    ->     COUNT(*) AS total_requests,
    ->     SUM(quantity) AS total_quantity,
    ->     AVG(quantity) AS average_quantity
    -> FROM request
    -> GROUP BY status
    -> ORDER BY status;
+-----------+----------------+----------------+------------------+
| status    | total_requests | total_quantity | average_quantity |
+-----------+----------------+----------------+------------------+
| pending   |              1 |           5000 |        5000.0000 |
| matched   |              6 |             38 |           6.3333 |
| delivered |              3 |             71 |          23.6667 |
+-----------+----------------+----------------+------------------+
3 rows in set (0.003 sec)

mysql>
```

7.2 UPDATE OPERATION

```
mysql> SET @pid = (SELECT MIN(person_id) FROM victim);
Query OK, 0 rows affected (0.004 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.000 sec)

mysql>
mysql> SELECT person_id, name, priority
    -> FROM victim
    -> WHERE person_id = @pid;
+-----------+------+----------+
| person_id | name | priority |
+-----------+------+----------+
|         1 | Asha | critical |
+-----------+------+----------+
1 row in set (0.001 sec)

mysql> UPDATE victim
    -> SET priority = IF(priority = 'high', 'low', 'high')
    -> WHERE person_id = @pid;
Query OK, 1 row affected (0.001 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT person_id, name, priority
    -> FROM victim
    -> WHERE person_id = @pid;
+-----------+------+----------+
| person_id | name | priority |
+-----------+------+----------+
|         1 | Asha | high     |
+-----------+------+----------+
1 row in set (0.000 sec)

mysql>
```

7.3 DELETE OPERATION

```
mysql> SELECT person_id, name, priority
    -> FROM victim
    -> WHERE person_id = @pid;
+-----------+------+----------+
| person_id | name | priority |
+-----------+------+----------+
|         1 | Asha | critical |
+-----------+------+----------+
1 row in set (0.000 sec)

mysql> SELECT * FROM route;
+----------+----------+------------+-------------+---------+
| route_id | depot_id | shelter_id | distance_km | status  |
+----------+----------+------------+-------------+---------+
|        1 |        1 |          1 |        3.50 | open    |
|        2 |        1 |          2 |        5.20 | open    |
|        3 |        2 |          2 |        7.80 | blocked |
|       17 |        2 |          3 |        8.00 | open    |
+----------+----------+------------+-------------+---------+
4 rows in set (0.000 sec)

mysql> SET @rid = (SELECT MIN(route_id) FROM route);
Query OK, 0 rows affected (0.000 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.000 sec)

mysql>
mysql> SELECT * FROM route WHERE route_id = @rid;
+----------+----------+------------+-------------+--------+
| route_id | depot_id | shelter_id | distance_km | status |
+----------+----------+------------+-------------+--------+
|        1 |        1 |          1 |        3.50 | open   |
+----------+----------+------------+-------------+--------+
1 row in set (0.001 sec)

mysql> DELETE FROM route
    -> WHERE route_id = @rid;
Query OK, 1 row affected (0.004 sec)

mysql> SELECT * FROM route
    -> WHERE route_id = @rid;
Empty set (0.001 sec)

mysql>
```

7.4 CORRELATED QUERY

```
mysql> SELECT * FROM route
    -> WHERE route_id = @rid;
+----------+----------+------------+-------------+--------+
| route_id | depot_id | shelter_id | distance_km | status |
+----------+----------+------------+-------------+--------+
|        1 |        1 |          1 |        3.50 | open   |
+----------+----------+------------+-------------+--------+
1 row in set (0.001 sec)

mysql> SELECT
    ->     v.person_id,
    ->     v.name,
    ->     v.priority
    -> FROM victim v
    -> WHERE EXISTS (
    ->     SELECT 1
    ->     FROM request r
    ->     WHERE r.person_id = v.person_id
    -> );
+-----------+-------+----------+
| person_id | name  | priority |
+-----------+-------+----------+
|         1 | Asha  | critical |
|         2 | Ravi  | high     |
|         3 | Meena | medium   |
|         4 | Karan | low      |
|         5 | Neha  | medium   |
|         8 | Manoj | high     |
+-----------+-------+----------+
6 rows in set (0.005 sec)

mysql>
```

7.5 NESTED QUERY

```
mysql> SELECT
    ->     request_id,
    ->     person_id,
    ->     resource_type,
    ->     quantity,
    ->     status
    -> FROM request
    -> WHERE quantity > (
    ->     SELECT AVG(quantity)
    ->     FROM request
    -> );
+------------+-----------+---------------+----------+---------+
| request_id | person_id | resource_type | quantity | status  |
+------------+-----------+---------------+----------+---------+
|         20 |         1 | water         |     5000 | pending |
+------------+-----------+---------------+----------+---------+
1 row in set (0.002 sec)

mysql>
```

8. STORED PROCEDURES, FUNCTIONS AND TRIGGERS

8.1 STORED PROCEDURES OR FUNCTIONS

```
(PES2UG24CS655@root) [disaster]> SELECT ROUTINE_TYPE, ROUTINE_NAME FROM information_schema.ROUTINES WHERE ROUTINE_SCHEMA='disaster';
+--------------+---------------------+
| ROUTINE_TYPE | ROUTINE_NAME        |
+--------------+---------------------+
| FUNCTION     | fn_distance_km      |
| FUNCTION     | fn_shelter_free     |
| PROCEDURE    | sp_complete_delivery |
| PROCEDURE    | sp_match_all_pending |
| PROCEDURE    | sp_match_request    |
+--------------+---------------------+
5 rows in set (0.01 sec)
```

Function execution:

1. fn_distance_km(lat1, lon1, lat2, lon2): returns the straight-line (Haversine) distance in km between two coordinates. Used to pick the nearest supply to a victim and the nearest vehicle to a supply, and to compute ETA.

```
(PES2UG24CS655@root) [disaster]> SELECT fn_distance_km(12.9716,77.5946,12.9352,77.6245) AS dist_km;
+-------------------+
| dist_km           |
+-------------------+
| 5.184651844857277 |
+-------------------+
1 row in set (0.00 sec)
```

2. fn_shelter_free(shelter_id): returns free capacity of a shelter (capacity - occupancy). Used in queries to show available space.

```
(PES2UG24CS655@root) [disaster]> SELECT shelter_id, name, fn_shelter_free(shelter_id) AS free_space FROM shelter;
+------------+--------------------------+------------+
| shelter_id | name                     | free_space |
+------------+--------------------------+------------+
|          1 | City School Shelter      |        150 |
|          2 | Stadium Shelter          |        380 |
|          3 | Temple Shelter           |        190 |
|          4 | Government School Shelter |        230 |
|          5 | Community Centre Shelter |        110 |
|          6 | Public Hall Shelter      |        130 |
|          7 | College Auditorium       |        250 |
|          8 | Municipal Shelter        |        200 |
|          9 | Sports Stadium North     |        300 |
|         10 | Village Community Hall   |        100 |
|         11 | Metro Community Shelter  |        195 |
|         12 | Relief Camp Alpha        |        250 |
|         13 | Relief Camp Beta         |        250 |
|         14 | Relief Camp Gamma        |        260 |
|         15 | Relief Camp Delta        |        125 |
|         16 | District Shelter         |        280 |
|         17 | Emergency Shelter        |        135 |
+------------+--------------------------+------------+
17 rows in set (0.00 sec)
```

Procedures execution:

1. sp_match_all_pending(): matches all pending requests in priority order (critical to low, then larger group, then earliest). It loops with a cursor calling sp_match_request and skips any that fail.

```
mysql> CALL sp_match_all_pending();
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> SELECT
    ->     m.match_id,
    ->     v.name,
    ->     v.priority,
    ->     r.resource_type,
    ->     r.quantity,
    ->     m.assigned_qty,
    ->     m.vehicle_id,
    ->     m.eta_minutes,
    ->     r.status
    -> FROM match_assignment m
    -> JOIN request r
    ->     ON m.request_id = r.request_id
    -> JOIN victim v
    ->     ON r.person_id = v.person_id
    -> ORDER BY
    ->     FIELD(v.priority,'critical','high','medium','low'),
    ->     m.match_id;
+----------+-------+----------+---------------+----------+-------------+------------+-------------+-----------+
| match_id | name  | priority | resource_type | quantity | assigned_qty | vehicle_id | eta_minutes | status    |
+----------+-------+----------+---------------+----------+-------------+------------+-------------+-----------+
|        1 | Asha  | critical | water         |       40 |          40 |          1 |           3 | matched   |
|        3 | Rekha | critical | water         |       10 |          10 |          4 |           3 | delivered |
|        2 | Ravi  | high     | medicine      |       10 |          10 |          2 |          14 | matched   |
|        4 | Meena | medium   | food          |       20 |          20 |         18 |           9 | matched   |
|        5 | Karan | low      | water         |       30 |          30 |         14 |          15 | matched   |
+----------+-------+----------+---------------+----------+-------------+------------+-------------+-----------+
5 rows in set (0.00 sec)
```

2. sp_complete_delivery(match_id): marks the request delivered (logged by trg_req_upd) and sets the vehicle back to available.

```
(PES2UG24CS655@root) [disaster]> CALL sp_complete_delivery(1);
Query OK, 1 row affected (0.02 sec)

(PES2UG24CS655@root) [disaster]> SELECT * FROM request_history
    -> WHERE request_id = (SELECT request_id FROM match_assignment WHERE match_id=1);
+------------+------------+------------+------------+---------------------+
| history_id | request_id | old_status | new_status | changed_at          |
+------------+------------+------------+------------+---------------------+
|          1 |          1 | NULL       | pending    | 2026-09-28 10:49:30 |
|          5 |          1 | pending    | matched    | 2026-09-28 10:50:35 |
|         22 |          1 | matched    | delivered  | 2026-10-02 21:58:33 |
+------------+------------+------------+------------+---------------------+
3 rows in set (0.00 sec)
```

3. sp_match_request(request_id): matches one request. It checks if the request is pending, finds the nearest supply with enough stock of that resource, finds the nearest available vehicle with enough capacity, computes ETA (at 40 km/h), and inserts into match_assignment. It raises an error if there is no supply or vehicle.

```
(PES2UG24CS655@root) [disaster]> CALL sp_match_request(@r);
Query OK, 1 row affected (0.02 sec)

(PES2UG24CS655@root) [disaster]> SELECT m.match_id, v.name, v.priority, r.resource_type, m.assigned_qty,
    ->        COALESCE(d.name,sh.name) AS collect_from, m.vehicle_id, m.eta_minutes, r.status
    -> FROM match_assignment m
    -> JOIN request r ON m.request_id=r.request_id
    -> JOIN victim v ON r.person_id=v.person_id
    -> JOIN supply s ON m.supply_id=s.supply_id
    -> LEFT JOIN depot d ON s.depot_id=d.depot_id
    -> LEFT JOIN shelter sh ON s.shelter_id=sh.shelter_id
    -> WHERE m.request_id=@r;
+----------+------------+----------+---------------+-------------+--------------+------------+-------------+---------+
| match_id | name       | priority | resource_type | assigned_qty | collect_from | vehicle_id | eta_minutes | status  |
+----------+------------+----------+---------------+-------------+--------------+------------+-------------+---------+
|        6 | Demo Victim | critical | water         |          10 | Central Depot |          1 |           2 | matched |
+----------+------------+----------+---------------+-------------+--------------+------------+-------------+---------+
1 row in set (0.01 sec)
```

```
-- ============================================================
-- FUNCTIONS
-- ============================================================

CREATE FUNCTION fn_distance_km(lat1 DOUBLE, lon1 DOUBLE, lat2 DOUBLE, lon2
DOUBLE)
RETURNS DOUBLE DETERMINISTIC
RETURN 6371*2*ASIN(SQRT(POW(SIN(RADIANS(lat2-lat1)/2),2)
  + COS(RADIANS(lat1))*COS(RADIANS(lat2))*POW(SIN(RADIANS(lon2-lon1)/2),2)));

CREATE FUNCTION fn_shelter_free(p_id INT) RETURNS INT DETERMINISTIC
RETURN (SELECT capacity-occupancy FROM shelter WHERE shelter_id=p_id);


-- ============================================================
-- PROCEDURES
-- ============================================================

DELIMITER //

-- Shared last step: nearest free vehicle to the supply point, ETA, then the INSERT
-- (trg_match_bi re-checks type and stock, trg_match_ai deducts stock and updates statuses).
CREATE PROCEDURE sp_assign_supply(IN p_req INT, IN p_supply INT)
BEGIN
  DECLARE v_qty INT;
  DECLARE v_lat DECIMAL(9,6); DECLARE v_lon DECIMAL(9,6);
  DECLARE v_slat DECIMAL(9,6); DECLARE v_slon DECIMAL(9,6);
  DECLARE v_veh INT; DECLARE v_vlat DECIMAL(9,6); DECLARE v_vlon DECIMAL(9,6);
  DECLARE v_dist DOUBLE; DECLARE v_eta INT;
  DECLARE CONTINUE HANDLER FOR NOT FOUND BEGIN END;

  SELECT r.quantity,v.latitude,v.longitude INTO v_qty,v_lat,v_lon
    FROM request r JOIN victim v ON r.person_id=v.person_id
    WHERE r.request_id=p_req;
```

```
  SELECT COALESCE(d.latitude,sh.latitude), COALESCE(d.longitude,sh.longitude) INTO
v_slat,v_slon
    FROM supply s
    LEFT JOIN depot d ON s.depot_id=d.depot_id
    LEFT JOIN shelter sh ON s.shelter_id=sh.shelter_id
    WHERE s.supply_id=p_supply;

  -- nearest available vehicle to the supply point
  SELECT vehicle_id,latitude,longitude INTO v_veh,v_vlat,v_vlon
    FROM vehicle
    WHERE status='available' AND capacity>=v_qty
    ORDER BY fn_distance_km(v_slat,v_slon,latitude,longitude) LIMIT 1;
  IF v_veh IS NULL THEN
    SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT='No vehicle available';
  END IF;

  SET v_dist = fn_distance_km(v_vlat,v_vlon,v_slat,v_slon) +
fn_distance_km(v_slat,v_slon,v_lat,v_lon);
  SET v_eta  = CEIL(v_dist/40*60);   -- assumes 40 km/h

  INSERT INTO match_assignment(request_id,supply_id,vehicle_id,assigned_qty,eta_minutes)
  VALUES (p_req,p_supply,v_veh,v_qty,v_eta);
END//

-- Automatic: nearest supply (depot or shelter) that holds enough stock.
CREATE PROCEDURE sp_match_request(IN p_req INT)
BEGIN
  DECLARE v_type VARCHAR(50); DECLARE v_qty INT; DECLARE v_status VARCHAR(20);
  DECLARE v_lat DECIMAL(9,6); DECLARE v_lon DECIMAL(9,6);
  DECLARE v_supply INT;
  DECLARE CONTINUE HANDLER FOR NOT FOUND BEGIN END;

  SELECT r.resource_type,r.quantity,r.status,v.latitude,v.longitude
    INTO v_type,v_qty,v_status,v_lat,v_lon
    FROM request r JOIN victim v ON r.person_id=v.person_id
    WHERE r.request_id=p_req;

  IF v_status IS NULL OR v_status <> 'pending' THEN
    SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT='Request not found or not pending';
  END IF;

  SELECT s.supply_id INTO v_supply
    FROM supply s
    LEFT JOIN depot d ON s.depot_id=d.depot_id
    LEFT JOIN shelter sh ON s.shelter_id=sh.shelter_id
    WHERE s.resource_type=v_type AND s.quantity>=v_qty
    ORDER BY
fn_distance_km(v_lat,v_lon,COALESCE(d.latitude,sh.latitude),COALESCE(d.longitude,sh.longitude)
)
```

```
    LIMIT 1;
  IF v_supply IS NULL THEN
    SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT='No supply available';
  END IF;

  CALL sp_assign_supply(p_req, v_supply);
END//

-- Chosen depot: fails with the requested and available amounts if it can't cover the request.
CREATE PROCEDURE sp_match_request_at_depot(IN p_req INT, IN p_depot INT)
BEGIN
  DECLARE v_type VARCHAR(50); DECLARE v_qty INT; DECLARE v_status VARCHAR(20);
  DECLARE v_supply INT; DECLARE v_have INT; DECLARE v_msg VARCHAR(128);
  DECLARE CONTINUE HANDLER FOR NOT FOUND BEGIN END;

  SELECT resource_type,quantity,status INTO v_type,v_qty,v_status
    FROM request WHERE request_id=p_req;
  IF v_status IS NULL OR v_status <> 'pending' THEN
    SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT='Request not found or not pending';
  END IF;

  SELECT supply_id,quantity INTO v_supply,v_have
    FROM supply WHERE depot_id=p_depot AND resource_type=v_type;
  IF v_supply IS NULL THEN
    SET v_msg = CONCAT('Depot has no ',v_type,' in stock');
    SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT=v_msg;
  END IF;
  IF v_have < v_qty THEN
    SET v_msg = CONCAT('Insufficient supply at depot: requested ',v_qty,', available ',v_have);
    SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT=v_msg;
  END IF;

  CALL sp_assign_supply(p_req, v_supply);
END//

CREATE PROCEDURE sp_match_all_pending()
BEGIN
  DECLARE done INT DEFAULT 0; DECLARE rid INT;
  DECLARE cur CURSOR FOR
    SELECT r.request_id FROM request r JOIN victim v ON r.person_id=v.person_id
    WHERE r.status='pending'
    ORDER BY FIELD(v.priority,'critical','high','medium','low'), v.group_size DESC, r.requested_at;
  DECLARE CONTINUE HANDLER FOR NOT FOUND SET done=1;
  OPEN cur;
  lp: LOOP
    FETCH cur INTO rid;
    IF done THEN LEAVE lp; END IF;
    BEGIN
```

```
    DECLARE CONTINUE HANDLER FOR SQLEXCEPTION BEGIN END;  -- skip unmatchable requests
    CALL sp_match_request(rid);
   END;
  END LOOP;
  CLOSE cur;
END//

CREATE PROCEDURE sp_complete_delivery(IN p_match INT)
BEGIN
  DECLARE v_req INT; DECLARE v_veh INT;
  SELECT request_id,vehicle_id INTO v_req,v_veh FROM match_assignment WHERE match_id=p_match;
  UPDATE request SET status='delivered' WHERE request_id=v_req;
  UPDATE vehicle SET status='available' WHERE vehicle_id=v_veh;
END//
DELIMITER ;
```

8.2 TRIGGERS

```
(PES2UG24CS655@root) [disaster]> SELECT TRIGGER_NAME, EVENT_MANIPULATION, EVENT_OBJECT_TABLE, ACTION_TIMING
    -> FROM information_schema.TRIGGERS WHERE TRIGGER_SCHEMA='disaster';
+-------------+--------------------+--------------------+--------------+
| TRIGGER_NAME | EVENT_MANIPULATION | EVENT_OBJECT_TABLE | ACTION_TIMING |
+-------------+--------------------+--------------------+--------------+
| trg_req_ins  | INSERT             | request            | AFTER        |
| trg_req_upd  | UPDATE             | request            | AFTER        |
| trg_match_bi | INSERT             | match_assignment   | BEFORE       |
| trg_match_ai | INSERT             | match_assignment   | AFTER        |
+-------------+--------------------+--------------------+--------------+
4 rows in set (0.00 sec)
```

Triggers execution:

1. trg_req_ins (AFTER INSERT on request): when a request is created, logs the first history row (NULL to pending).

```
mysql> SELECT *
    -> FROM request_history
    -> ORDER BY history_id;
+------------+------------+------------+------------+---------------------+
| history_id | request_id | old_status | new_status | changed_at          |
+------------+------------+------------+------------+---------------------+
|          1 |          1 | NULL       | pending    | 2026-09-28 10:49:30 |
|          2 |          2 | NULL       | pending    | 2026-09-28 10:49:30 |
|          3 |          3 | NULL       | pending    | 2026-09-28 10:49:30 |
|          4 |          4 | NULL       | pending    | 2026-09-28 10:49:30 |
|          5 |          1 | pending    | matched    | 2026-09-28 10:50:35 |
|          6 |          2 | pending    | matched    | 2026-09-28 10:50:35 |
|          7 |          5 | NULL       | pending    | 2026-09-28 11:29:43 |
|          8 |          5 | pending    | matched    | 2026-09-28 11:30:09 |
|          9 |          5 | matched    | delivered  | 2026-09-28 11:30:58 |
+------------+------------+------------+------------+---------------------+
9 rows in set (0.00 sec)
```

2. trg_req_upd (AFTER UPDATE on request): when the status changes, logs old to new status with a timestamp. This is the audit trail.

```
(PES2UG24CS655@root) [disaster]> UPDATE request SET status='delivered' WHERE request_id=2;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

(PES2UG24CS655@root) [disaster]> SELECT * FROM request_history WHERE request_id=2;
+------------+------------+------------+------------+---------------------+
| history_id | request_id | old_status | new_status | changed_at          |
+------------+------------+------------+------------+---------------------+
|          2 |          2 | NULL       | pending    | 2026-09-28 10:49:30 |
|          6 |          2 | pending    | matched    | 2026-09-28 10:50:35 |
|         23 |          2 | matched    | delivered  | 2026-10-02 22:00:58 |
+------------+------------+------------+------------+---------------------+
3 rows in set (0.00 sec)
```

3. trg_match_bi (BEFORE INSERT on match_assignment): validates before a match is stored. Rejects it if the supply's resource type differs from the request's, or if stock is less than the assigned quantity.

```
(PES2UG24CS655@root) [disaster]> INSERT INTO match_assignment(request_id,supply_id,vehicle_id,assigned_qty,eta_minutes)
    -> VALUES (2,1,NULL,10,10);
ERROR 1644 (45000): Resource type mismatch
```

4. trg_match_ai (AFTER INSERT on match_assignment): once a match is stored, it deducts the assigned quantity from supply, sets the request to matched, and sets the vehicle to en_route. The request update in turn fires trg_req_upd.

```
(PES2UG24CS655@root) [disaster]> SELECT m.match_id, m.assigned_qty, s.supply_id, s.quantity AS supply_left,
    ->        r.status AS req_status, v.status AS veh_status
    -> FROM match_assignment m
    -> JOIN supply s ON m.supply_id=s.supply_id
    -> JOIN request r ON m.request_id=r.request_id
    -> LEFT JOIN vehicle v ON m.vehicle_id=v.vehicle_id;
+----------+-------------+-----------+-------------+------------+------------+
| match_id | assigned_qty | supply_id | supply_left | req_status | veh_status |
+----------+-------------+-----------+-------------+------------+------------+
|        1 |          40 |         1 |          50 | delivered  | available  |
|        2 |          10 |         4 |          20 | delivered  | en_route   |
|        3 |          10 |         1 |          50 | delivered  | available  |
|        4 |          20 |        19 |          55 | matched    | en_route   |
|        5 |          30 |         3 |          30 | matched    | en_route   |
+----------+-------------+-----------+-------------+------------+------------+
5 rows in set (0.00 sec)
```

```
-- ============================================================
-- TRIGGERS
-- ============================================================

DELIMITER //
CREATE TRIGGER trg_req_ins AFTER INSERT ON request FOR EACH ROW
BEGIN
  INSERT INTO request_history(request_id,old_status,new_status)
  VALUES (NEW.request_id,NULL,NEW.status);
END//

CREATE TRIGGER trg_req_upd AFTER UPDATE ON request FOR EACH ROW
BEGIN
  IF OLD.status <> NEW.status THEN
```

```
    INSERT INTO request_history(request_id,old_status,new_status)
    VALUES (NEW.request_id,OLD.status,NEW.status);
  END IF;
END//

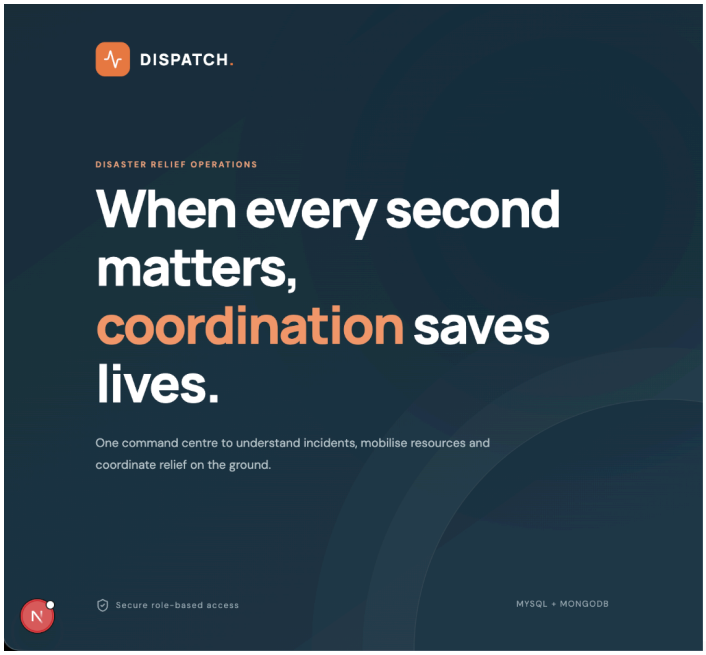
CREATE TRIGGER trg_match_bi BEFORE INSERT ON match_assignment FOR EACH ROW
BEGIN
  DECLARE v_have INT; DECLARE v_t1 VARCHAR(50); DECLARE v_t2 VARCHAR(50);
  SELECT quantity,resource_type INTO v_have,v_t1 FROM supply WHERE
supply_id=NEW.supply_id;
  SELECT resource_type INTO v_t2 FROM request WHERE request_id=NEW.request_id;
  IF v_t1 <> v_t2 THEN
    SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT='Resource type mismatch';
  END IF;
  IF v_have < NEW.assigned_qty THEN
    SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT='Insufficient supply';
  END IF;
END//

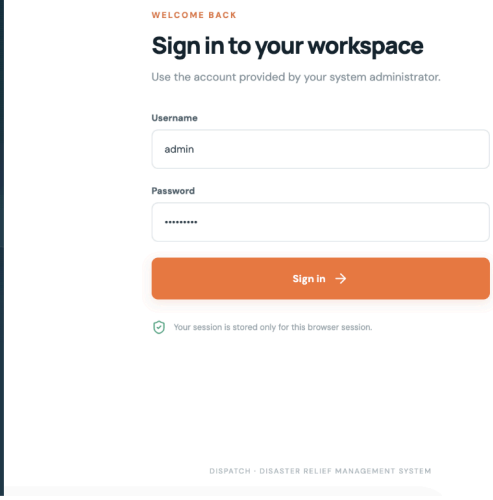
CREATE TRIGGER trg_match_ai AFTER INSERT ON match_assignment FOR EACH ROW
BEGIN
  UPDATE supply SET quantity=quantity-NEW.assigned_qty WHERE supply_id=NEW.supply_id;
  UPDATE request SET status='matched' WHERE request_id=NEW.request_id;
  IF NEW.vehicle_id IS NOT NULL THEN
    UPDATE vehicle SET status='en_route' WHERE vehicle_id=NEW.vehicle_id;
  END IF;
END//

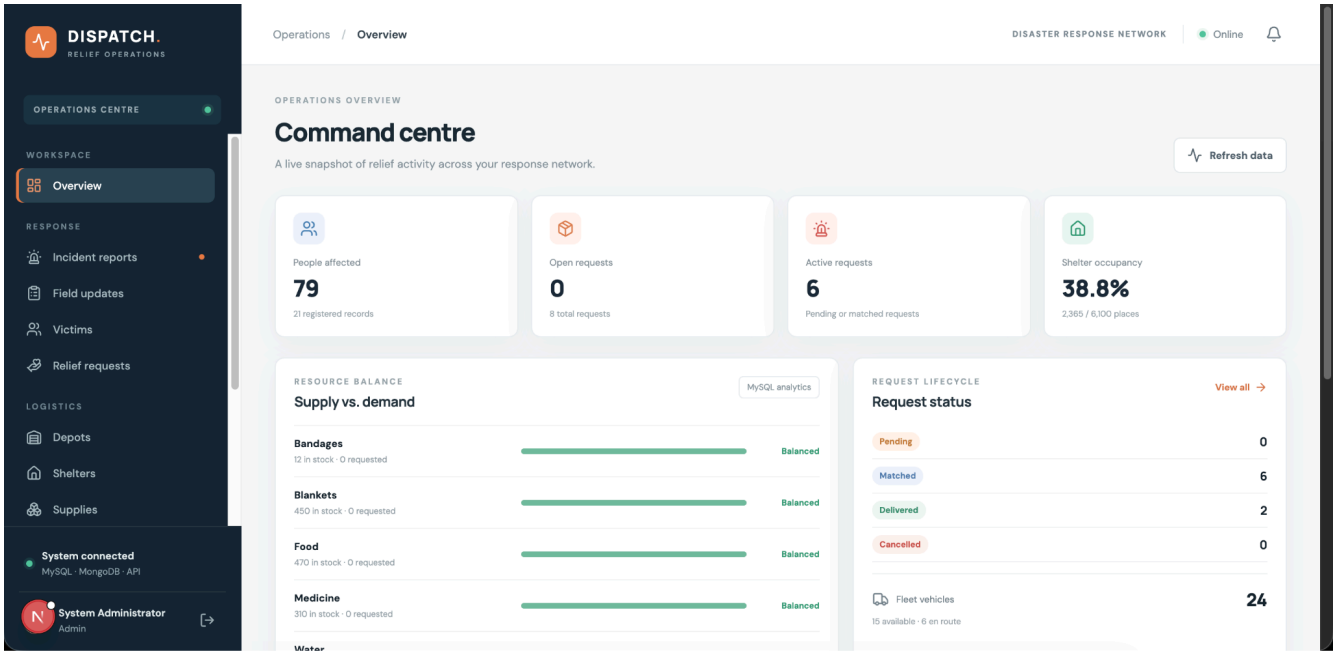
DELIMITER ;
```

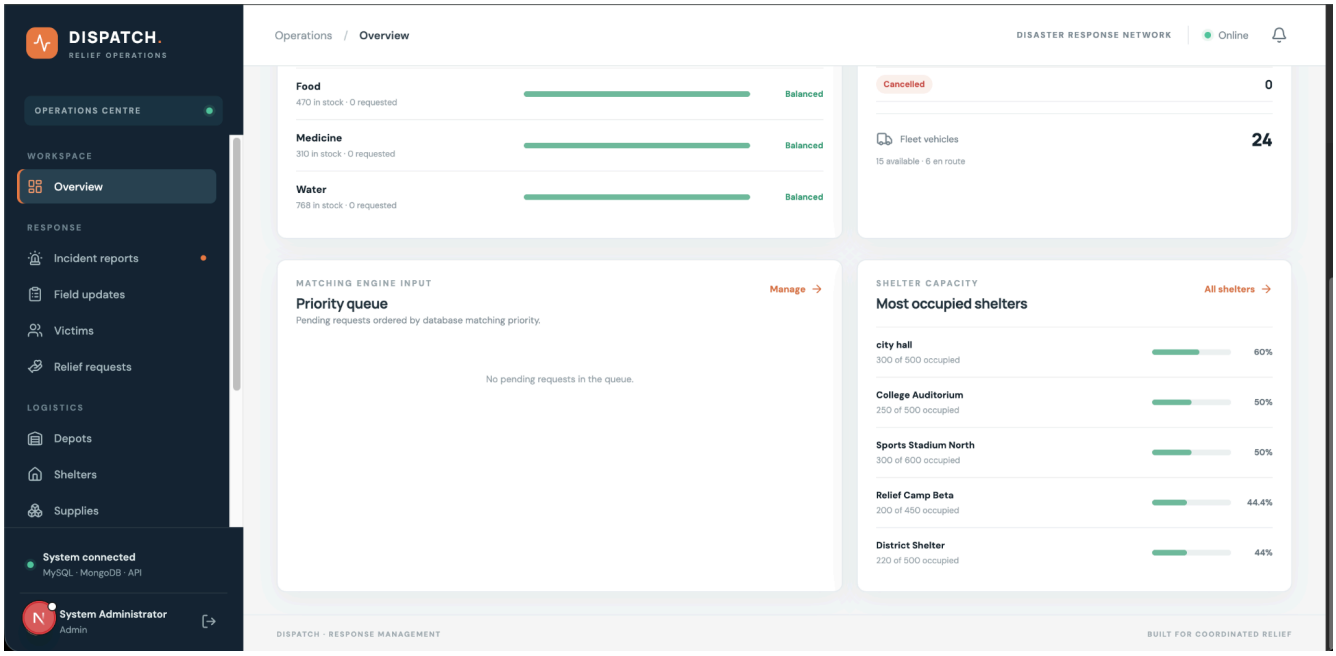
9. FRONT END DEVELOPMENT

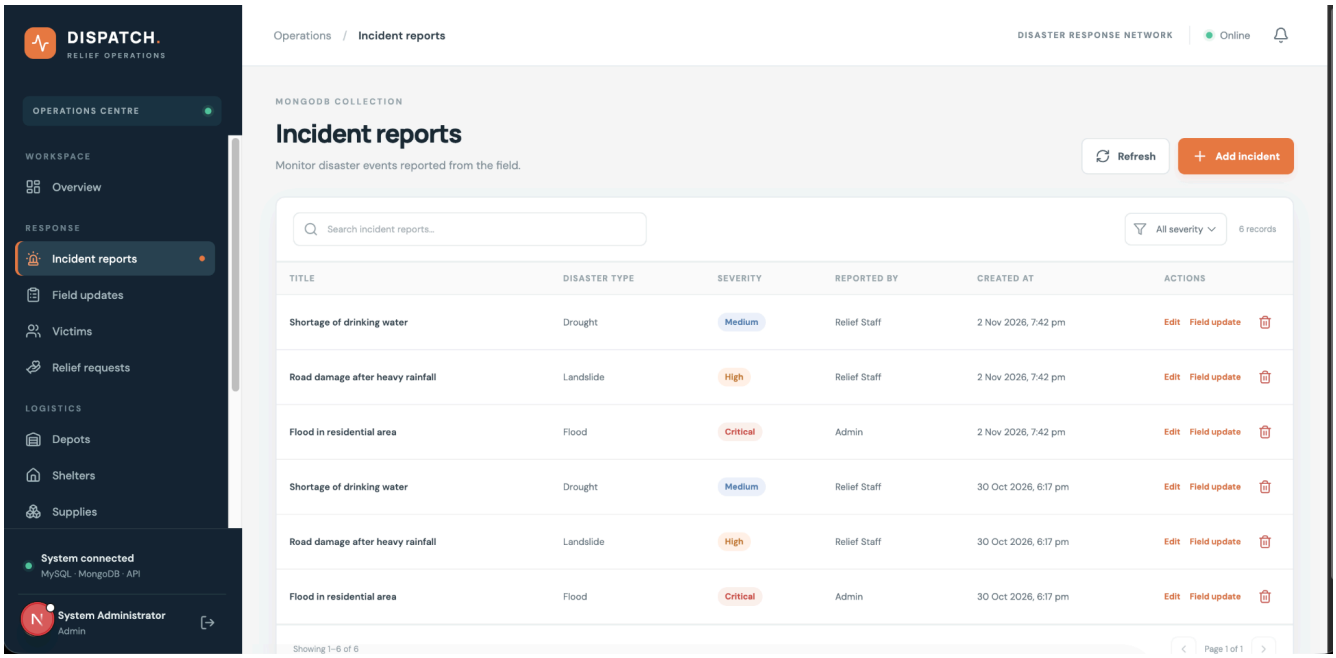
screenshots:

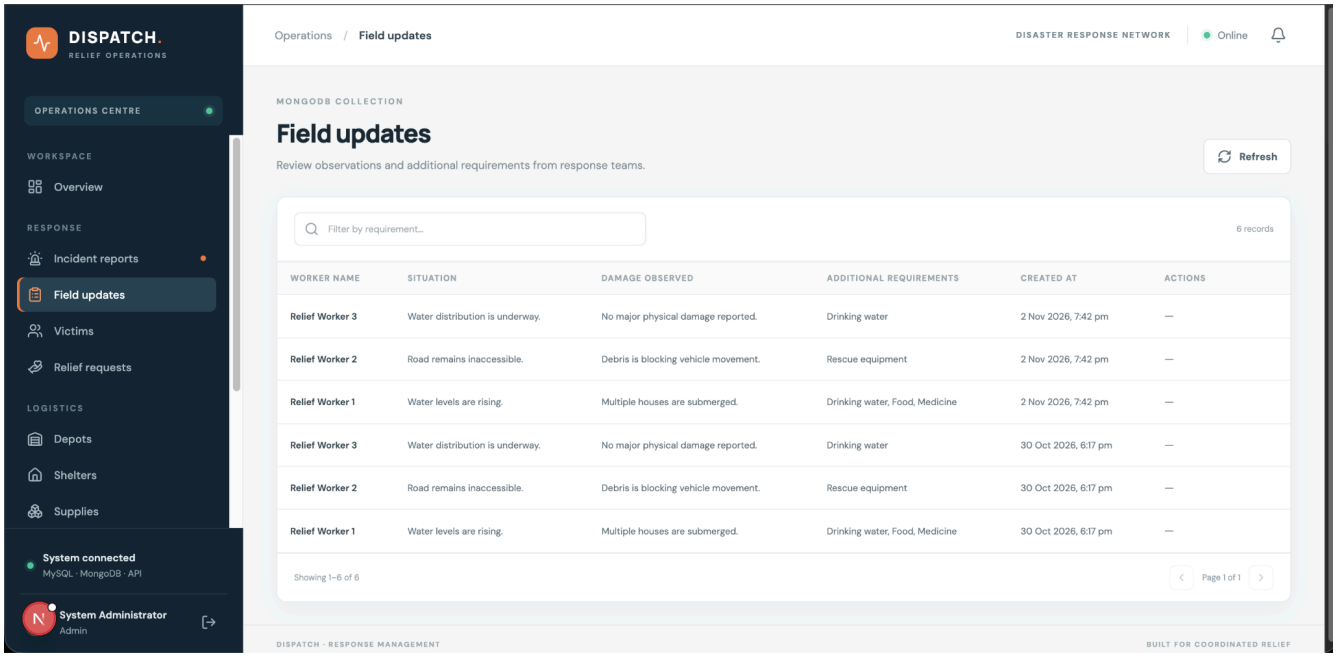


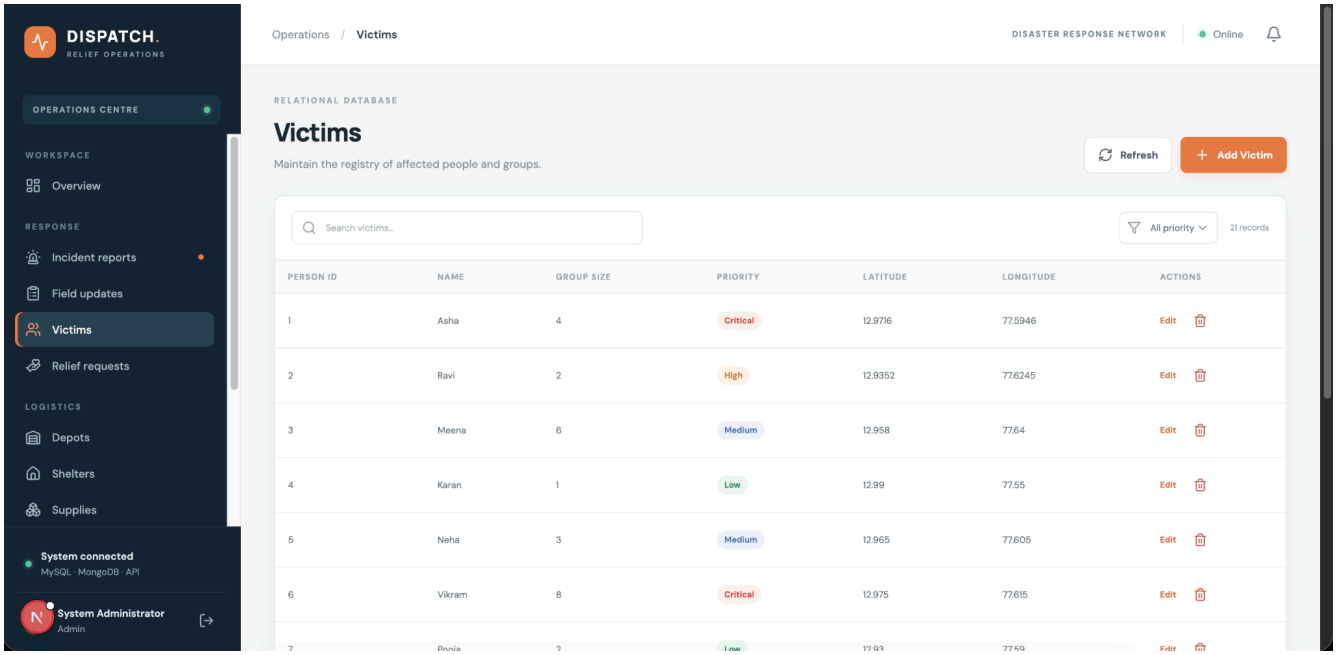


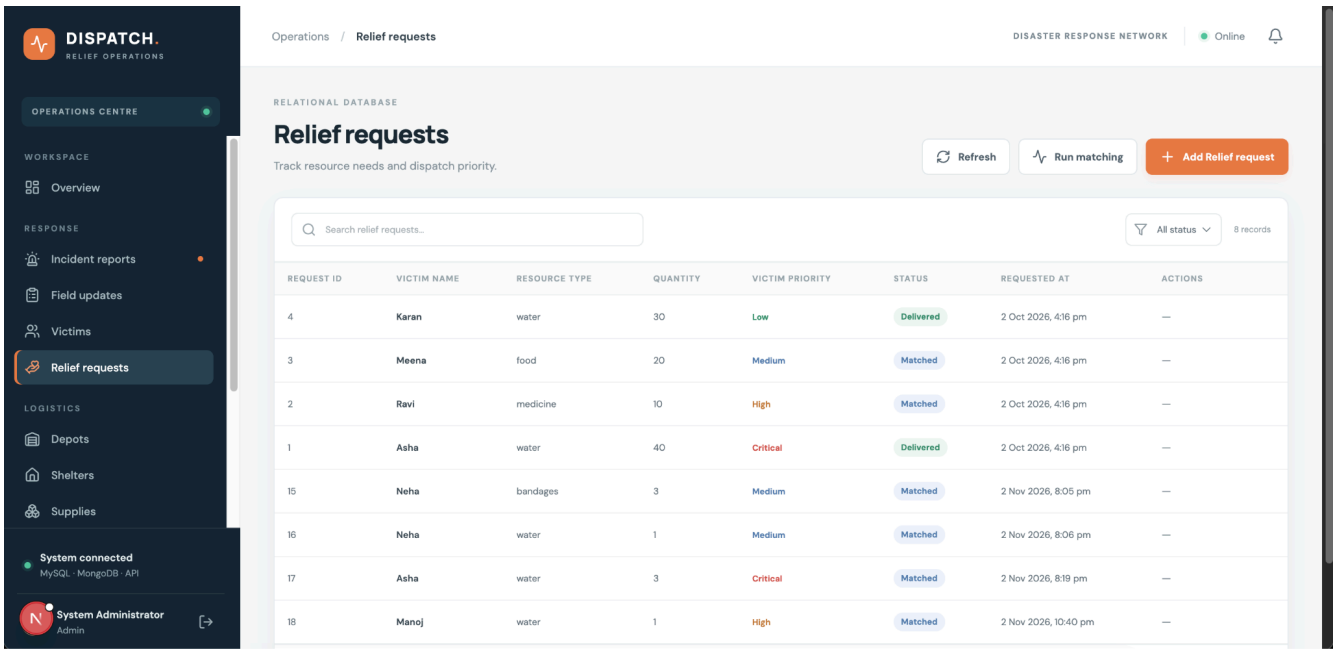


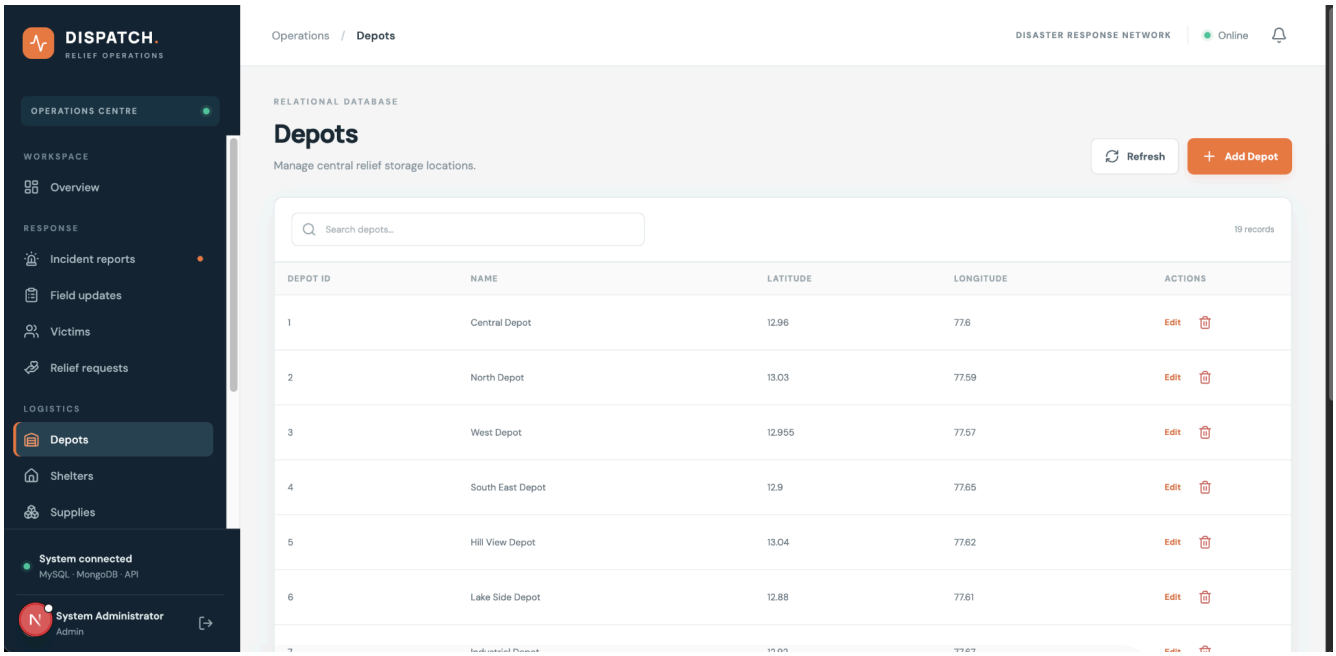

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Food
470 in stock · 0 requested
Balanced
Medicine
310 in stock · 0 requested
Balanced
Water
768 in stock · 0 requested
Balanced
Cancelled
0
Fleet vehicles
15 available · 6 en route
24
MATCHING ENGINE INPUT
Priority queue
Pending requests ordered by database matching priority.
Manage
No pending requests in the queue.
SHELTER CAPACITY
Most occupied shelters
All shelters
city hall
300 of 500 occupied
60%
College Auditorium
250 of 500 occupied
50%
Sports Stadium North
300 of 600 occupied
50%
Relief Camp Beta
200 of 450 occupied
44.4%
District Shelter
220 of 500 occupied
44%
DISPATCH · RESPONSE MANAGEMENT
BUILT FOR COORDINATED RELIEF

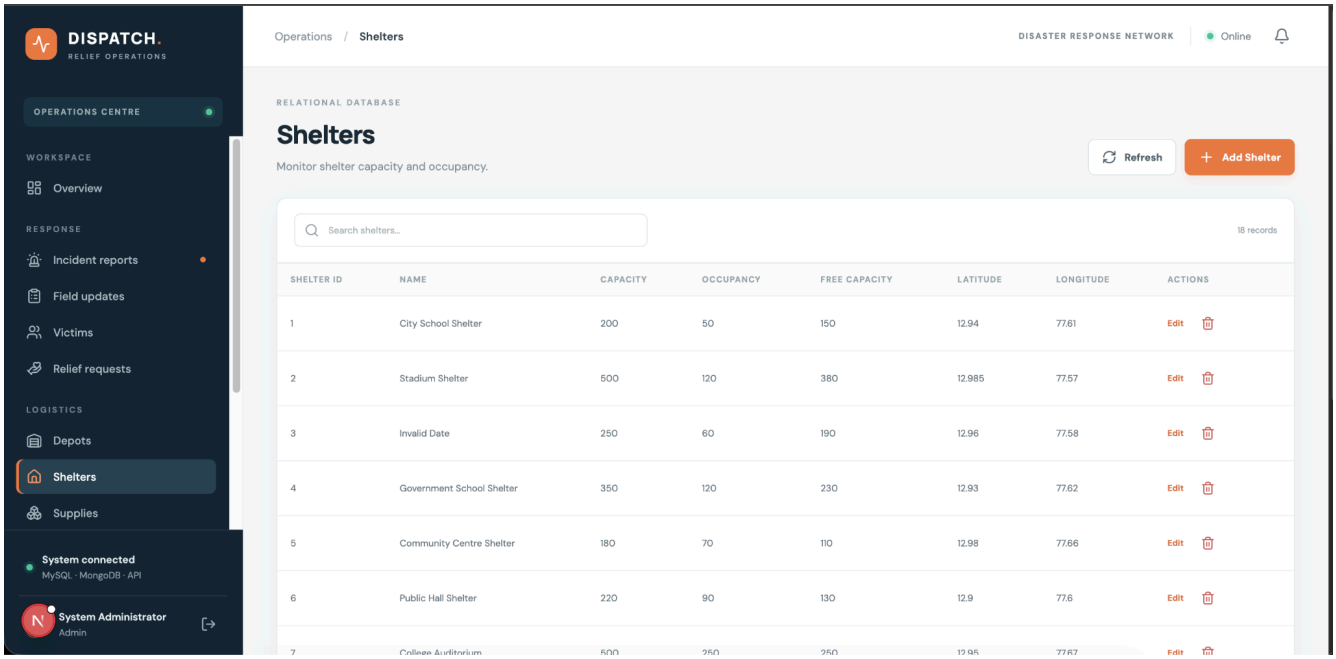

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MONGODB COLLECTION
Incident reports
Monitor disaster events reported from the field.
Refresh
Add incident
Search incident reports...
All severity
6 records
TITLE
DISASTER TYPE
SEVERITY
REPORTED BY
CREATED AT
ACTIONS
Shortage of drinking water
Drought
Medium
Relief Staff
2 Nov 2026, 7:42 pm
Edit
Field update
Road damage after heavy rainfall
Landslide
High
Relief Staff
2 Nov 2026, 7:42 pm
Edit
Field update
Flood in residential area
Flood
Critical
Admin
2 Nov 2026, 7:42 pm
Edit
Field update
Shortage of drinking water
Drought
Medium
Relief Staff
30 Oct 2026, 6:17 pm
Edit
Field update
Road damage after heavy rainfall
Landslide
High
Relief Staff
30 Oct 2026, 6:17 pm
Edit
Field update
Flood in residential area
Flood
Critical
Admin
30 Oct 2026, 6:17 pm
Edit
Field update
Showing 1–6 of 6
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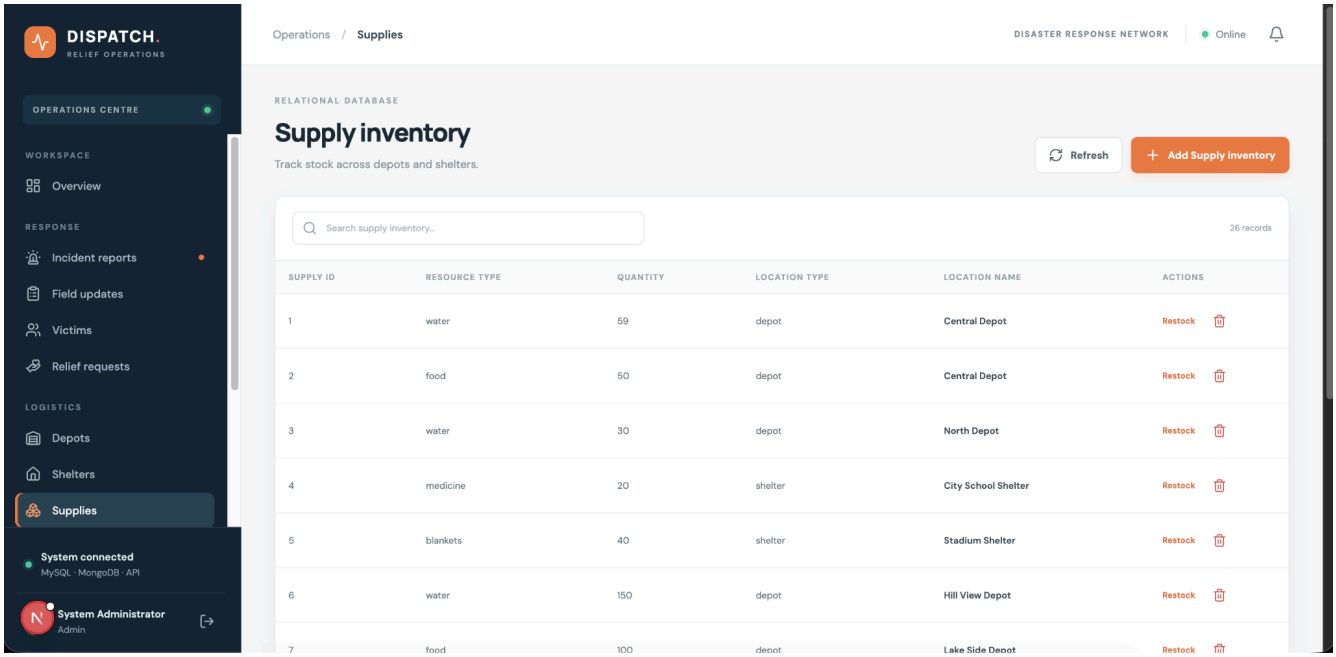

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MONGODB COLLECTION
Field updates
Review observations and additional requirements from response teams.
Refresh
Filter by requirement...
6 records
WORKER NAME
SITUATION
DAMAGE OBSERVED
ADDITIONAL REQUIREMENTS
CREATED AT
ACTIONS
Relief Worker 3
Water distribution is underway.
No major physical damage reported.
Drinking water
2 Nov 2026, 7:42 pm
Relief Worker 2
Road remains inaccessible.
Debris is blocking vehicle movement.
Rescue equipment
2 Nov 2026, 7:42 pm
Relief Worker 1
Water levels are rising.
Multiple houses are submerged.
Drinking water, Food, Medicine
2 Nov 2026, 7:42 pm
Relief Worker 3
Water distribution is underway.
No major physical damage reported.
Drinking water
30 Oct 2026, 6:17 pm
Relief Worker 2
Road remains inaccessible.
Debris is blocking vehicle movement.
Rescue equipment
30 Oct 2026, 6:17 pm
Relief Worker 1
Water levels are rising.
Multiple houses are submerged.
Drinking water, Food, Medicine
30 Oct 2026, 6:17 pm
Showing 1–6 of 6
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DISPATCH · RESPONSE MANAGEMENT
BUILT FOR COORDINATED RELIEF

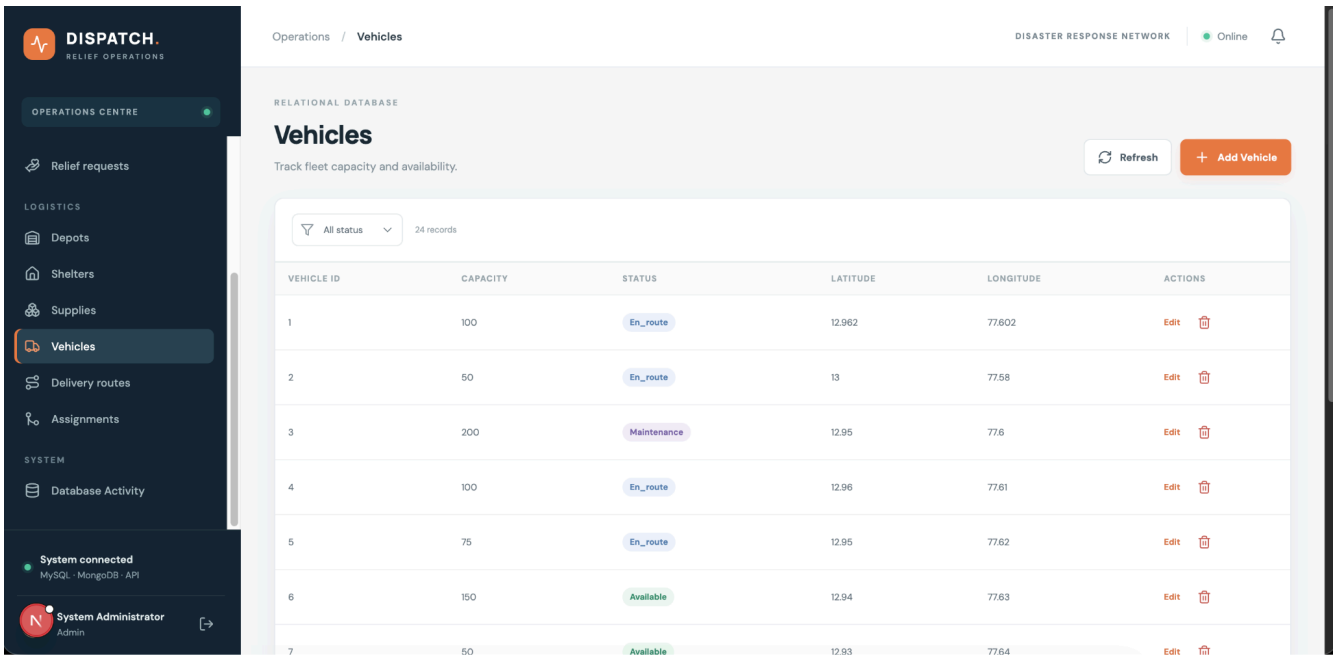

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RELATIONAL DATABASE
Victims
Maintain the registry of affected people and groups.
Refresh
Add Victim
Search victims...
All priority
21 records
PERSON ID
NAME
GROUP SIZE
PRIORITY
LATITUDE
LONGITUDE
ACTIONS
1
Asha
4
Critical
12.9716
77.5946
Edit
2
Ravi
2
High
12.9352
77.6245
Edit
3
Meena
6
Medium
12.958
77.64
Edit
4
Karan
1
Low
12.99
77.55
Edit
5
Neha
3
Medium
12.965
77.605
Edit
6
Vikram
8
Critical
12.975
77.615
Edit

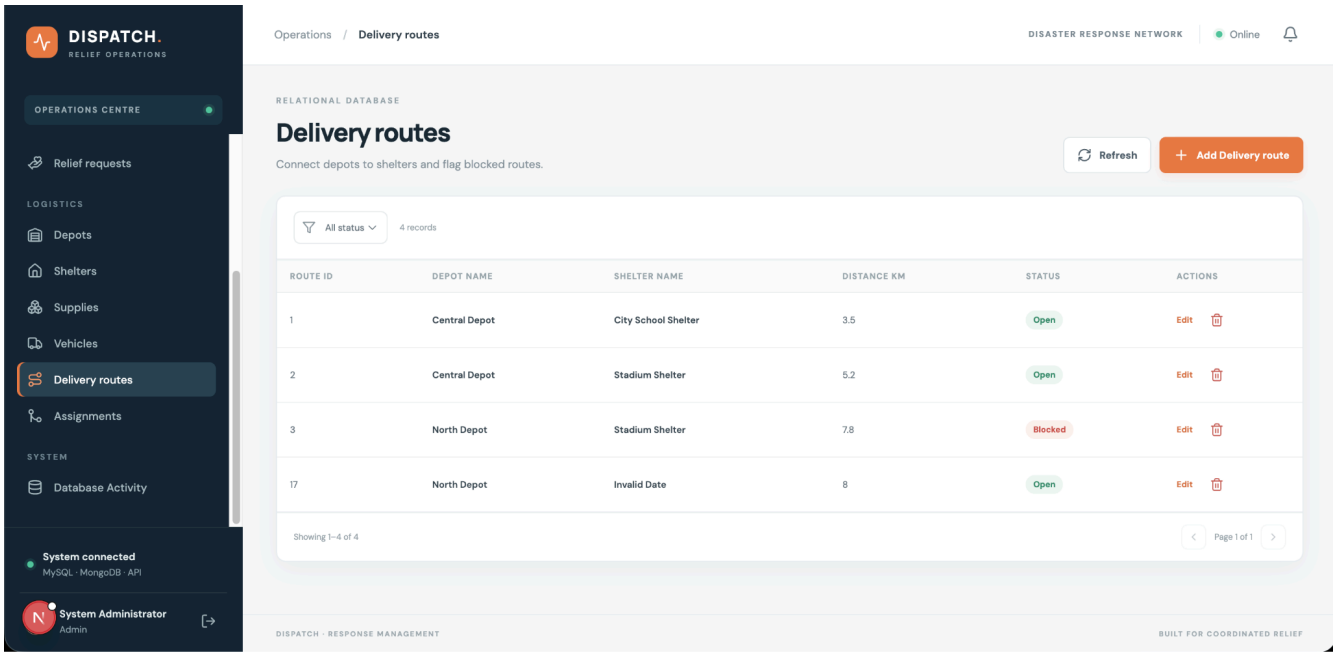

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DISASTER RESPONSE NETWORK
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RELATIONAL DATABASE
Relief requests
Track resource needs and dispatch priority.
Refresh
Run matching
Add Relief request
Search relief requests...
All status
8 records
REQUEST ID
VICTIM NAME
RESOURCE TYPE
QUANTITY
VICTIM PRIORITY
STATUS
REQUESTED AT
ACTIONS
4 Karan water 30 Low Delivered 2 Oct 2026, 4:16 pm —
3 Meena food 20 Medium Matched 2 Oct 2026, 4:16 pm —
2 Ravi medicine 10 High Matched 2 Oct 2026, 4:16 pm —
1 Asha water 40 Critical Delivered 2 Oct 2026, 4:16 pm —
15 Neha bandages 3 Medium Matched 2 Nov 2026, 8:05 pm —
16 Neha water 1 Medium Matched 2 Nov 2026, 8:06 pm —
17 Asha water 3 Critical Matched 2 Nov 2026, 8:19 pm —
18 Manoj water 1 High Matched 2 Nov 2026, 10:40 pm —

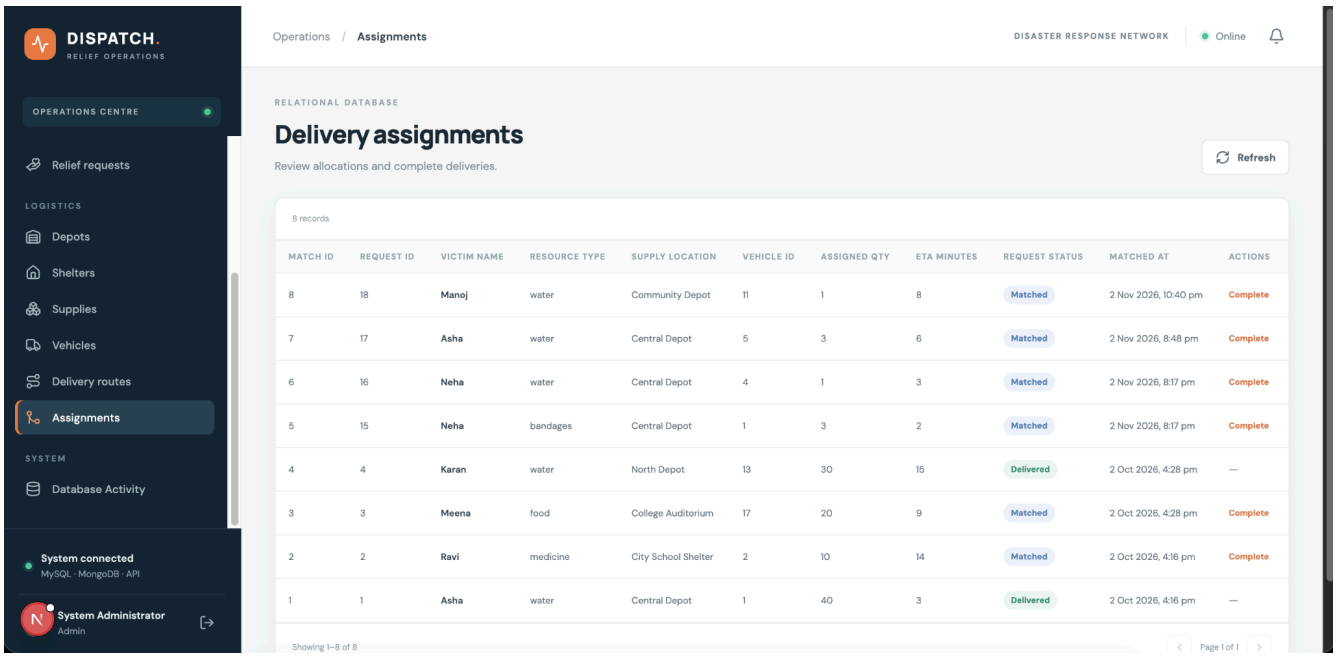

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RELATIONAL DATABASE
Depots
Manage central relief storage locations.
Refresh
Add Depot
Search depots...
19 records
DEPOT ID
NAME
LATITUDE
LONGITUDE
ACTIONS
1 Central Depot 12.96 77.6 Edit
2 North Depot 13.03 77.59 Edit
3 West Depot 12.955 77.57 Edit
4 South East Depot 12.9 77.65 Edit
5 Hill View Depot 13.04 77.62 Edit
6 Lake Side Depot 12.88 77.61 Edit

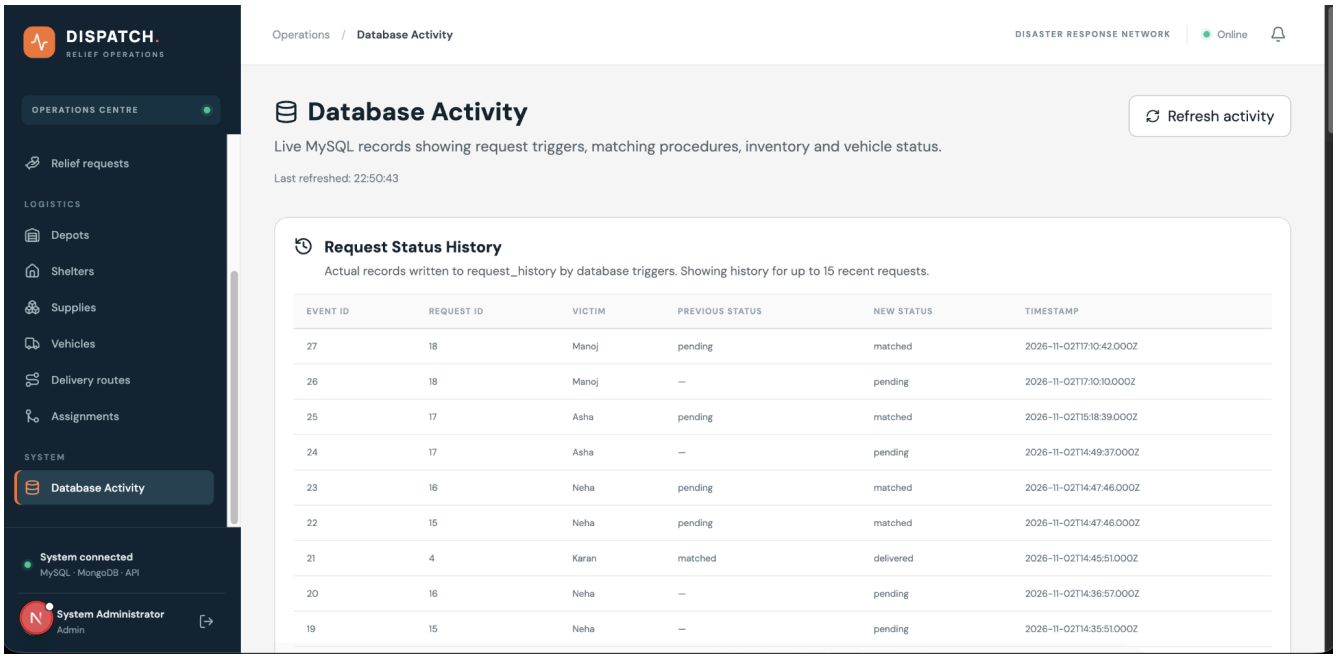

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Operations / Shelters
DISASTER RESPONSE NETWORK
Online
RELATIONAL DATABASE
Shelters
Monitor shelter capacity and occupancy.
Refresh
Add Shelter
Search shelters...
18 records
SHELTER ID
NAME
CAPACITY
OCCUPANCY
FREE CAPACITY
LATITUDE
LONGITUDE
ACTIONS
1 City School Shelter 200 50 150 12.94 77.61 Edit
2 Stadium Shelter 500 120 380 12.985 77.57 Edit
3 Invalid Date 250 60 190 12.96 77.58 Edit
4 Government School Shelter 350 120 230 12.93 77.62 Edit
5 Community Centre Shelter 180 70 110 12.98 77.66 Edit
6 Public Hall Shelter 220 90 130 12.9 77.6 Edit
7 College Auditorium 500 250 250 12.95 77.67 Edit


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RELATIONAL DATABASE
Supply inventory
Track stock across depots and shelters.
Refresh
Add Supply inventory
Search supply inventory...
26 records
SUPPLY ID
RESOURCE TYPE
QUANTITY
LOCATION TYPE
LOCATION NAME
ACTIONS
1 water 59 depot Central Depot Restock
2 food 50 depot Central Depot Restock
3 water 30 depot North Depot Restock
4 medicine 20 shelter City School Shelter Restock
5 blankets 40 shelter Stadium Shelter Restock
6 water 150 depot Hill View Depot Restock
7 food 100 depot Lake Side Depot Restock


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Vehicles
Track fleet capacity and availability.
Refresh
Add Vehicle
All status
24 records
VEHICLE ID
CAPACITY
STATUS
LATITUDE
LONGITUDE
ACTIONS
1 100 En_route 12.962 77.602 Edit
2 50 En_route 13 77.58 Edit
3 200 Maintenance 12.95 77.6 Edit
4 100 En_route 12.96 77.61 Edit
5 75 En_route 12.95 77.62 Edit
6 150 Available 12.94 77.63 Edit
7 50 Available 12.93 77.64 Edit


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Delivery routes
Connect depots to shelters and flag blocked routes.
Refresh
Add Delivery route
All status
4 records
ROUTE ID
DEPOT NAME
SHELTER NAME
DISTANCE KM
STATUS
ACTIONS
1 Central Depot City School Shelter 3.5 Open Edit
2 Central Depot Stadium Shelter 5.2 Open Edit
3 North Depot Stadium Shelter 7.8 Blocked Edit
17 North Depot Invalid Date 8 Open Edit
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GitHub Link:

https://github.com/rosa36-x/disaster-management-system